Improving Precipitation Phase Classification in Mountain Hydrology: A Machine Learning Framework Using Citizen Science and Gridded Meteorological Data

Emma Golub,[a] Nayoung Hur,[a] Meghan Collins,[b] and the Mountain Rain or Snow Citizen Scientists[c]

[a] *Lynker, Louisville, CO*
[b] *Desert Research Institute, Reno, NV*
[c] *Mountain Rain or Snow project,* ***[TODO: coordinating institution]***

ABSTRACT: Precipitation phase partitioning is a leading source of uncertainty in mountain hydrology, where rain–snow misclassification degrades streamflow forecasting and water-resource management. Conventional methods that rely on sparse station networks and near-surface thermodynamic thresholds are limited near freezing, where rain and snow distributions overlap. We explore whether citizen-science observations and gridded data can better predict phase there. Our framework assimilates crowdsourced phase reports from the Mountain Rain or Snow (MRoS) network with station meteorology and satellite-derived probability of liquid precipitation, interpolates every source to an hourly 1-km grid, and trains a calibrated machine learning model on held-out MRoS labels across California's Sierra Nevada and Colorado's Rocky Mountains from October 2022 to May 2026. Because mixed precipitation is difficult to classify as a distinct phase, we instead compare observer-reported mixed events against the model's own region of uncertainty, which flags where mixed phase is likely. We benchmark against nine standard partitioning methods and quantify significance with a paired cluster bootstrap. Test AUROC was 0.965 in both domains (95% CI 0.957–0.972 in California; 0.942–0.982 in Colorado). Against the best benchmark, the model gained 11.1 percentage points of accuracy in California (9.0–13.4) and 16.5 points near freezing (12.2–21.1). Colorado's gains were smaller on accuracy (3.6 points, 0.2–6.9) but clearer on macro F1 (0.096, 0.042–0.152). Ablations that retrain with inputs withheld trace these gains to the assimilated predictors, above all the crowdsourced surfaces.

SIGNIFICANCE STATEMENT: Predicting whether mountain precipitation falls as rain or snow is essential for water-supply and flood forecasting. However, the process is challenging and prone to errors near freezing temperatures, where sparse weather stations and simple temperature-based partitioning methods struggle to correctly classify precipitation phase. We show that combining crowdsourced phase reports from volunteer observers with station, satellite, and terrain data on a common 1-km grid substantially improves rain–snow prediction across California and Colorado, especially near freezing.

1. Introduction

Determining whether cold-season precipitation falls as rain or snow is a critical yet challenging area of research in mountain hydrology. Seasonal mountain snowpack supplies water to more than one-sixth of the global population (Barnett et al. 2005) and governs streamflow timing and magnitude. In the western United States, 53% of total runoff originates as snowmelt, rising to roughly 70% in mountainous terrain (Li et al. 2017). Precipitation phase largely determines how that water is delivered. Snow accumulates through the winter and is released gradually during melt, whereas rain runs off within hours, so a single storm can either recharge water supply or trigger flooding depending on its phase. An error in phase classification propagates quickly downstream, leading to inaccurate snowpack estimates, runoff timing, flood magnitude estimation, and weather forecasts issued to mountain communities and backcountry users (Harpold et al. 2017; Jennings and Molotch 2019). Morevoer, warming temperatures have been responsible in shifting cold-season precipitation from snow to rain and increasing rain-on-snow flood risk across western North America (Musselman et al. 2018), both of which contribute to higher uncertainty in water supply planning and hazards forecasting.

Phase remains difficult to classify accurately, especially near freezing. Precipitation gauges located in mountainous terrain are geographically sparse relative to the heterogeneous conditions they are intended to record. Few stations report phase at all, and those that do often sit where cold-air pooling, wind exposure, and steep elevation gradients decouple recorded estimates from true conditions in their surroundings (Lundquist et al. 2008; Minder et al. 2011; Wayand et al. 2016). Precipitation phase also resists a clean definition: near freezing, hydrometeors may arrive as graupel, ice pellets, freezing rain, or wet mixtures that share a range of surface air temperatures (Section d). This known physical ambiguity has made classifying mixed precipitation separately from rain or snow a predictive challenge and significant source of error.

Conventional phase partitioning models struggle to accurately predict phase. Most land surface, hydrologic, and snow models partition rain from snow with a single near-surface air-temperature threshold or a smooth transition around one (Ding et al. 2014; Dai 2008). This approach

Corresponding author: Emma Golub, emmaa.golub@gmail.com

measurably alters simulated accumulation and melt (Jennings and Molotch 2019) without translating to regions where transitional phase temperatures can vary by more than 4 °C (Jennings et al. 2018). Wet-bulb temperature supplements phase classification by capturing the cooling of hydrometeors falling through unsaturated air (Harder and Pomeroy 2013; Sims and Liu 2015; Wang et al. 2019), but no surface measurement resolves melting and refreezing aloft, so errors concentrate near freezing (Feiccabrino et al. 2015). Increasing model complexity does not necessarily fix the issue either; Jennings et al. (2025) found that fine-tuning machine learning models marginally improves phase predictions that rely on simple partitioning rules and near-surface meteorology.

Addressing this predictive accuracy ceiling requires exploring new model inputs. Ground-based phase reports at transition-zone elevations can supplement synoptic weather stations, which are sparse and typically biased toward lower elevations and airports. Crowdsourcing helps fill atmospheric observations where instruments are absent (Muller et al. 2015), and the Mountain Rain or Snow (MRoS) project explores this through collecting timestamped, geolocated visual reports from citizen scientist volunteers across mountain regions (Arienzo et al. 2021). Such reports have helped in characterizing rain–snow from mixed conditions and have been used in benchmarking phase partitioning methods (Jennings et al. 2023, 2025). To our knowledge, their value as a predictor—assimilated alongside station and satellite data into a spatially continuous, probabilistic field—has yet to be explored.

This study explores how to overcome existing limitations in phase prediction. The following three research questions frame this work:

1. Do assimilated predictors (i.e., interpolated crowdsourced phase observations, interpolated station meteorology, and satellite-derived probability of liquid precipitation (pLP)) improve rain–snow accuracy beyond what near-surface thermodynamics alone can supply?

2. Does this data assimilation approach enhance predictive phase accuracy near freezing where conventional threshold partititioning methods fail?

3. Can mixed precipitation, which is usually discarded or forced into an ill-defined third class, exhibit meaningful signal in a model's region of predictive uncertainty?

The framework we propose combines four complementary sources of data: crowdsourced phase reports from MRoS, hourly station meteorology, satellite pLP from GPM IMERG (Huffman et al. 2020), and topographical elevation. The combination of these data types makes up in strength for what each lacks on its own. Stations are typically accurate but provide sparse data points. Satellite products capture a precipitating column but with coarse resolution. Crowdsourced observations are similarly sparse and can be biased to one storm but can report phase that actually reached the ground where stations are absent. The station meteorology and crowdsourced observations are interpolated into their own surfaces, which are then assimilated with resampled IMERG pLP and elevation onto a common hourly 1-km grid.

We train an XGBoost classifier (Chen and Guestrin 2016) on held-out MRoS phase labels and evaluate it in two climatologically distinct ranges, California's Sierra Nevada and Colorado's Rocky Mountains, which differ in precipitation pattern, terrain, and observer density. After bootstrapping with clustered resampling to quantify significance and produce confidence intervals for each prediction, we benchmark against the standard partitioning methods surveyed by Jennings et al. (2025).

2. Study domains and period

Two mountain domains are processed independently through an identical pipeline: the California Sierra Nevada / Lake Tahoe region (UTM Zone 11N, EPSG:26911) and the Colorado Rocky Mountains (UTM Zone 13N, EPSG:32613). Each area of interest is bounded by a polygon over mountain terrain, the study period spans 1 October 2022 to 1 May 2026 (excluding June, July, and August as the summer non-observation season), and all datasets are harmonized to a common 1-km analysis grid whose coordinates and elevations derive from a digital elevation model (DEM). The domains contrast usefully (Figure 1): California has much wider covariation of air temperature with elevation, atmospheric-river-driven storms, and a comparatively balanced observation mix, whereas Colorado's continental snow climate is strongly snow-dominant with sparser rain.

3. Data

a. Crowdsourced phase observations

MRoS citizen scientists submit timestamped, geolocated reports of precipitation phase (rain, snow, or mixed) during storm events (Arienzo et al. 2021). Volunteers who have opted into storm text alerts, issued when forecast surface temperatures fall in the rain–snow uncertainty range of roughly −2 to 10 °C, open a browser-based app during precipitation and record what phase is currently falling; the device supplies geolocation and timestamp automatically. The protocol operationalizes the phase definitions of Section 1: observers report what is falling rather than what has accumulated, classify graupel as snow and freezing rain as rain, exclude hail and true sleet, and reserve the mixed label for genuinely ambiguous or rapidly transitioning precipitation. That last instruction matters, because it makes the mixed label an explicit report of observer uncertainty

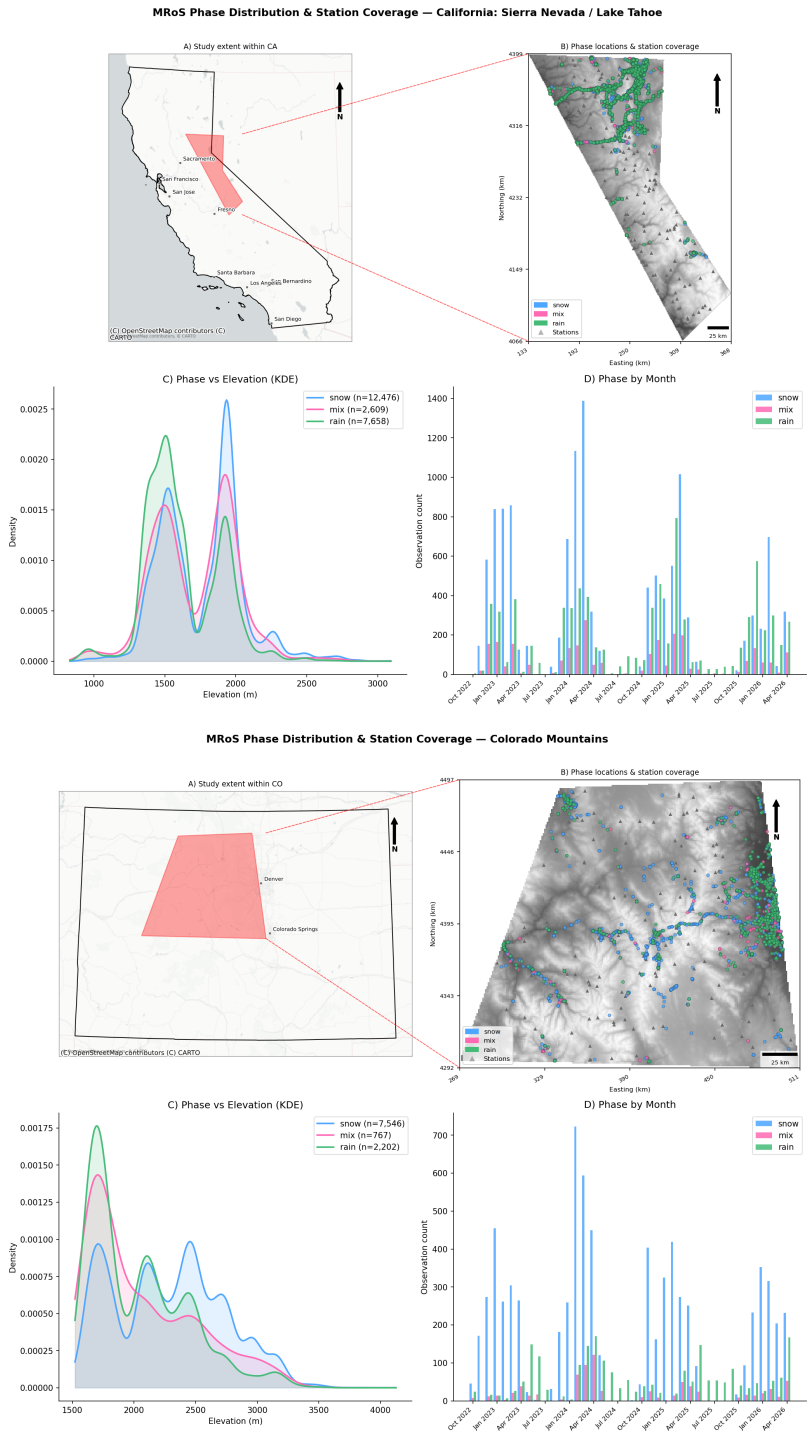


FIG. 1. Study domains and observation distribution for (a, top) California's Sierra Nevada and Lake Tahoe and (b, bottom) Colorado's Rocky Mountains. Within each block: (A) study extent (red) within the state; (B) MRoS reports over shaded relief, colored by reported phase, with assimilated surface stations as grey triangles; (C) kernel density of report elevation by phase; and (D) monthly report counts by phase. Panel C is density-normalized within each phase, so the curves compare shape rather than volume; phase totals appear in the legend (California 12,476 snow, 2,609 mix, 7,658 rain; Colorado 7,546 snow, 767 mix, 2,202 rain). California's reports are bimodal in elevation, rain-dominated near 1500 m and snow-dominated near 1950 m, whereas Colorado's peak near 1700 m with secondary snow modes above; this is the elevation–temperature covariation invoked later to explain California's non-monotonic skill structure. Two sampling caveats apply throughout: reports cluster along road corridors and near population centers, so panel B reflects observer access as much as precipitation frequency, and panel D counts are participation-driven with near-total summer gaps, so they are not storm climatology.

rather than a claim about a distinct hydrometeor type. Neither temperature nor intensity is entered by the observer, so each record reduces to a geotagged, timestamped phase label.

Each submission is processed with the `rainOrSnowTools` package (Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024), which tags it with USGS 3DEP elevation and with modeled surface meteorology from stations within a one-degree radius (requiring at least five stations) after standardized range and consistency checks. Quality assurance follows the procedures established for the network (Arienzo et al. 2021; Jennings et al. 2023): only real-time reports are retained, since ancillary fields are matched to each report's exact time and location, and observations lacking nearby station support or failing physical plausibility checks are removed. After this, the rain–snow observation set comprises 15,337 observations in California (10,735 train / 2,301 validation / 2,301 test; 64% snow) and 3,492 in Colorado (2,444 / 524 / 524; 82% snow). Observer-reported mixed events constitute a further 12.2% of California test observations (320 of 2,621) and 10.9% of Colorado's (64 of 588); they are held aside from binary training and used for uncertainty-band evaluation (Section d).

b. Station meteorology

Hourly surface observations are collected from four public networks (HADS, LCD, WCC, and MADIS) via the `rainOrSnowTools` access layer (Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024). The retained variables are air temperature, dewpoint, and relative humidity, from which wet-bulb temperature is derived during assimilation (Section a). These stations provide the thermodynamic environment that is interpolated to the analysis grid.

c. Satellite probability of liquid precipitation

Half-hourly GPM IMERG probability-of-liquid-precipitation (pLP) fields (Huffman et al. 2020) are retrieved from NASA GES DISC via OPeNDAP, subset server-side to each domain, and merged across the 48 half-hourly timesteps into daily gridded tables. pLP is an independent, satellite-derived estimate of precipitation phase (only rain or snow) from passive microwave and infrared retrievals and later becomes a key model predictor: it carries information about the precipitating column that near-surface measurements cannot typically supply.

d. Reanalysis climate fields (explored, not used)

We explored daily PRISM rasters (Daly et al. 2008) as a spatially complete backdrop to the sparse station network, resampling five variables from 800 m to the 1-km grid. PRISM regridded cleanly in space, however its daily resolution is too coarse for an hourly phase product. Expanding it to hourly required imposing a diurnal temperature curve, and that synthetic sub-daily structure did not improve validation skill over the station-interpolated fields while risking bias near freezing. We therefore excluded it. Coarser global reanalyses (ERA5, MERRA-2) were similarly rejected because their 30–50-km resolutions cannot resolve the kilometer-scale elevational gradients that control phase here.

e. Terrain

Raw 10-m DEMs from the USGS 3DEP / National Map are clipped to each domain, reprojected to the regional UTM CRS, and resampled to 1 km by average pooling. The resulting DEM defines the analysis grid and supplies the elevation field used for lapse-rate detrending, kriging drift, and as a model predictor.
Table 1 lists the final predictor set, which comprises eight fields once the three MRoS surfaces are counted separately (discussed in (Section b)). Relative humidity is interpolated and used to derive dewpoint and wet-bulb temperature, but it is not itself supplied to the model (Section d).

4. Methods

The methodology divides into two pipelines that are run identically per domain. A data-preparation pipeline (Sections a–b) fuses station meteorology, external gridded products, and MRoS reports, then interpolates them onto the common analysis grid, producing gridded predictor fields and leakage-safe MRoS phase-probability predictors. A modeling pipeline (Sections c–3) trains and calibrates the binary classifier, derives the mixed-phase uncertainty band, and evaluates the result through ablation, benchmarking, and a clustered bootstrap. Figure 2 summarizes the full data-assimilation and modeling architecture.

a. Data assimilation and quality control

An assimilation stage cleans the station record, derives consistent meteorology, and harmonizes every source to the DEM grid before interpolation. Missing thermodynamic variables are reconstructed from physical relationships: saturation vapor pressure follows Magnus–Tetens, dewpoint and relative humidity are interchanged to fill whichever is missing, and wet-bulb temperature, the most physically informative single predictor of phase, is estimated with Stull's formula (Stull 2011), refined by an iterative psychrometric solution and constrained between dewpoint and air temperature. Where a station reports air temperature without co-located humidity, dewpoint and relative humidity are gap-filled at the station using lapse-rate-adjusted inverse-distance weighting (IDW) from

TABLE 1. Model predictors and their sources. All fields are harmonized to the hourly 1-km analysis grid.

Predictor	Source	Native resolution
Air temperature	HADS/LCD/WCC/MADIS stations, interpolated	point, hourly
Dewpoint temperature	Stations, interpolated	point, hourly
Wet-bulb temperature	Derived (Stull 2011)	—
Elevation	USGS 3DEP DEM	10 m
IMERG pLP	GPM IMERG (Huffman et al. 2020)	0.1°, half-hourly
MRoS $p_{\text{snow}}, p_{\text{mix}}, p_{\text{rain}}$	LOOCV-interpolated MRoS reports	point, hourly

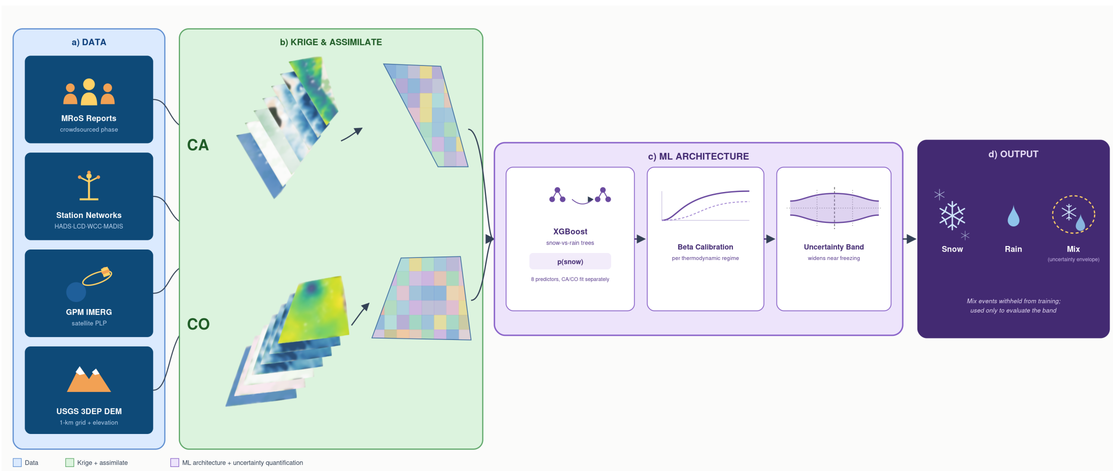


FIG. 2. Data-assimilation and machine-learning architecture. (a) Inputs: crowdsourced MRoS phase reports, surface station networks (HADS, LCD, WCC, MADIS), GPM IMERG probability of liquid precipitation (pLP), and the USGS 3DEP elevation model resampled to the 1-km grid. (b) Station observations and MRoS reports are kriged to that grid per domain and timestamp, producing the predictor stack sampled at each report location. (c) An XGBoost snow-versus-rain classifier is fit separately per domain on eight predictors, its scores are beta-calibrated, and the wet-bulb-conditioned band of Equation 2 is applied. (d) Output is snow, rain, or an uncertainty flag. Two design choices govern how the rest of the paper should be read. The classifier is binary throughout, and the uncertainty flag is a post-hoc rule applied to the calibrated score, not a third trained class. And mixed events are withheld from training entirely, serving only as held-out ground truth for evaluating band placement, which is why mix capture is a diagnostic rather than a classification score.

neighboring stations, following the `model_meteo` procedure of `rainOrSnowTools` (Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024).

Station observations then pass through quality control: duplicates are dropped, hard physical cutoffs remove impossible readings, a per-station interquartile-range filter removes outliers, spike and stuck-sensor detection nulls implausible runs, and cross-variable checks reconcile temperature with humidity. Survivors are aggregated to one record per station per hour. A dynamic lapse rate is estimated at each timestep from the station temperature–elevation relationship, letting the vertical structure vary hour to hour, and DEM elevation is appended to every point dataset so all sources share an elevation reference. IMERG is brought to the hourly 1-km grid by bilinear resampling, and all retained fields are synchronized on a common UTC axis and written to hourly NetCDF stacks.

b. Spatial interpolation

All point observations are interpolated to the 1-km analysis grid by kriging, the best linear unbiased predictor for a spatial field (Cressie 1993; Goovaerts 1997). Kriging estimates an unsampled location as $\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i)$, with weights λ_i that minimize estimation variance subject to unbiasedness. Those weights are set by the spatial dependence structure, summarized by the empirical semivariogram over observation pairs separated by lag $\mathbf{h}$,

$$\hat{\gamma}(\mathbf{h}) = \frac{1}{2\,|N(\mathbf{h})|} \sum_{(i,j)\in N(\mathbf{h})} \left[Z(\mathbf{s}_i) - Z(\mathbf{s}_j)\right]^2, \qquad (1)$$

to which a spherical model was fitted operationally. Because hourly observations are sparse, we pool the empirical semivariogram across time before fitting, which stabilizes the covariance estimate during thinly observed hours.

Each variable is kriged in the form matched to its physical nature. Air and dewpoint temperature carry a strong elevation trend, so they are detrended against station elevation with the hourly lapse rate, interpolated by ordinary kriging on the residuals, and retrended with DEM elevations, a regression-kriging that preserves physical vertical structure (Hengl et al. 2007). Relative humidity is interpolated by universal kriging with DEM elevation as an external drift term (Wackernagel 2003), so its terrain dependence enters the trend directly.

Precipitation phase is nominal, so encoding snow, mix, or rain ordinally and interpolating would impose artificial arithmetic (i.e., mix would represent a meaningless average of snow and rain). We therefore convert phase to one-hot indicators, krige each class independently by indicator kriging (Journel 1983), then clip to non-negative values and renormalize so the three probabilities sum to one, yielding hourly surfaces p_{snow}, p_{mix}, p_{rain}. Figure 3 shows the full set at one storm timestamp per domain.

(i) Leakage prevention. MRoS observations serve both as training labels and—through the interpolated phase surfaces—as candidate predictors. Reading an observation's predictor from a full-grid surface built with that same observation would let the label leak into the feature. To break the dependence we compute the MRoS phase predictors by pointwise spatial leave-one-out cross-validation (LOOCV): each report is removed from its hour, the remaining reports in that hour are re-kriged by indicator kriging to the held-out location under the same minimum-neighbor requirement, and the resulting triplet is clipped, renormalized, and stored as the leakage-safe predictor. What remains is the phase implied by other observers nearby, which is legitimate spatial context available at inference time rather than the target label itself. Full-grid surfaces built from all observations are retained separately for mapping and gridded prediction, never as training features. This mirrors the leave-one-out logic used for unbiased local estimates in geostatistical cross-validation (Goovaerts 1997).

c. Machine learning model

We frame phase prediction as binary classification and train an XGBoost model (Chen and Guestrin 2016) with a logistic objective on rain–snow observations only. We refer to the resulting eight-predictor configuration as "MRoS-XGB" throughout for simpler mention. The output is the probability of snow, with $p(\text{rain}) = 1 - p(\text{snow})$. Gridded predictors are sampled at each MRoS observation location from the hourly NetCDF stacks, giving eight predictors: air, dewpoint, and wet-bulb temperature, elevation, IMERG pLP, and the three leakage-safe LOOCV MRoS phase probabilities. Data are split 70/15/15 into train/validation/test, stratified on observed phase class with a fixed random seed, and early stopping is applied on the validation subset (up to 2,000 rounds, patience 50; selected iterations 323 in CA and 513 in CO).

Two discarded alternatives motivate this structure. A multiclass version trained snow/mix/rain directly and was abandoned because mixed precipitation does not form a separable cluster in feature space (Section d). A second version kept binary training but used spatially blocked splits with Platt-scaled calibration, and was abandoned because blocking left calibration sets too small to fit stably across thermodynamic regimes. The randomized stratified split adopted here instead preserves the distribution of meteorological states across subsets, matching the evaluation design of Jennings et al. (2025), and defers the residual spatial correlation to the inference stage (Section 3). Its known cost is that observations from the same storm may appear in both train and test, so test performance may be mildly optimistic relative to a novel season.

(i) Class imbalance. Both training sets skew toward snow (64% in California, 82% in Colorado), and the skew is not simply a matter of counts. The MRoS network favors higher-elevation, snowier sites, and those sites are also colder, drier, and carry higher LOOCV snow probabilities, so the class imbalance is entangled with the feature space. Left uncorrected it produces a model that favors snow, surfacing as snow recall exceeding rain recall and doing most damage near the transition, where a missed rain call is the error that matters for runoff timing. The standard remedy, frequency-inverse weighting, balances the effective counts but does not target the recall asymmetry itself, and because the geographic and thermodynamic skews compound, it can over- or under-correct. We therefore tune XGBoost's `scale_pos_weight` against the quantity of interest; a probe model is trained across a grid of candidate weights and the value minimizing the absolute snow–rain recall gap on the validation set is selected (0.50 for California, 0.15 for the more snow-dominant Colorado). Both fall below one, downweighting the majority snow class, and the resulting recalls balance to within 0.03 and 0.06 respectively (Section a). Because class weighting shifts the raw score distribution, calibration follows it rather than preceding it.

(ii) Probability calibration. A classifier can rank cases correctly while reporting probabilities that do not verify. Gradient-boosted trees are particularly prone to this, since each boosting round drives training loss down by pushing scores toward 0 and 1, and the resulting overconfidence is amplified under class imbalance and by the weighting above. That matters here because downstream applications need a $p(\text{snow})$ that means what it reports, and because the uncertainty envelope discussed further in Section d is an interval in probability space whose placement is meaningful only if the probabilities are.

We therefore fit a post-hoc map from raw score to calibrated probability. Two features of the problem shape how. First, discrimination near 0 °C wet-bulb differs fundamentally from discrimination far from freezing, because near

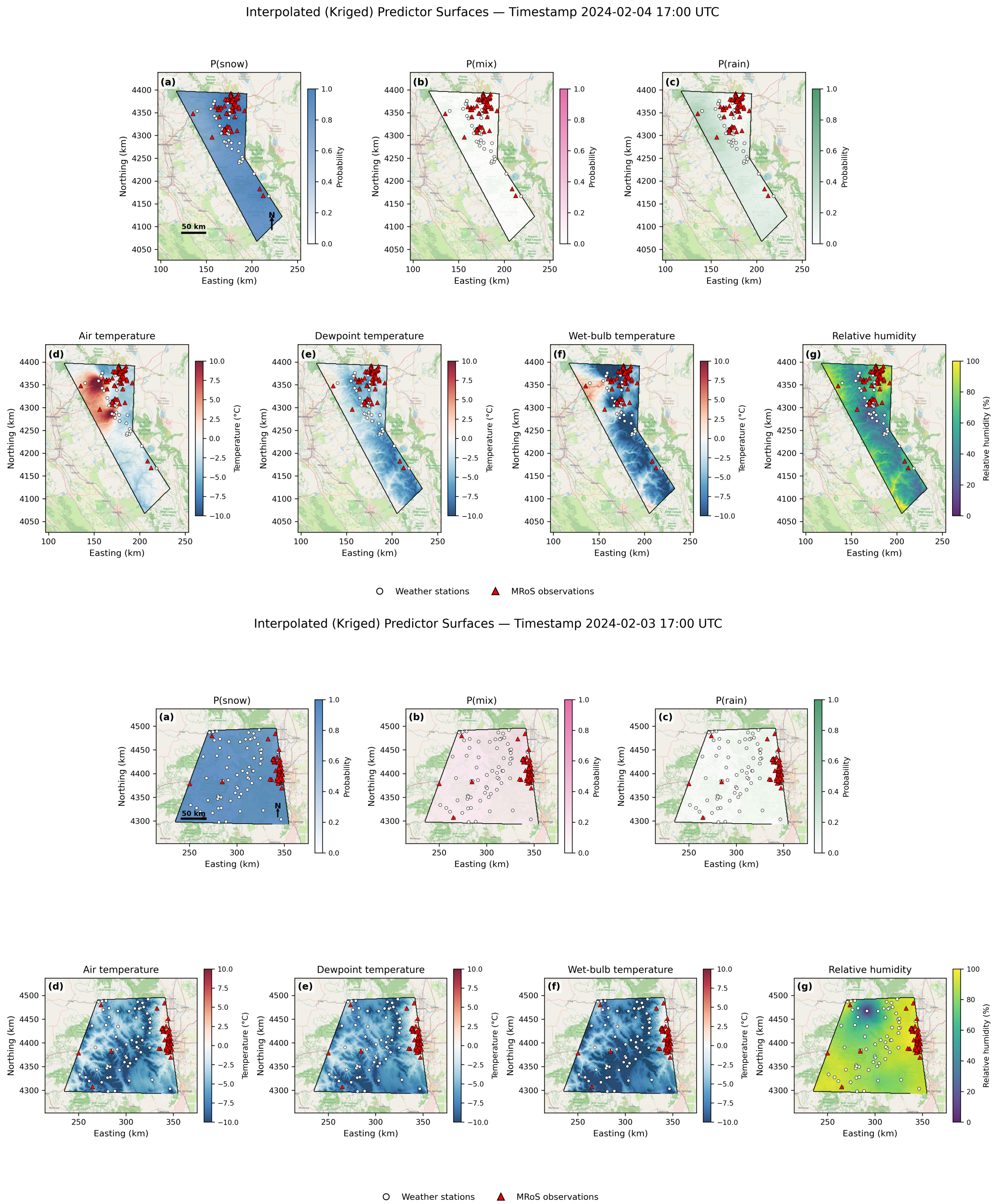


Fig. 3. Kriged predictor surfaces on the 1-km analysis grid for (a, top) California at 2024-02-04 17:00 UTC and (b, bottom) Colorado at 2024-02-03 17:00 UTC, two independent storm timestamps. Within each block, panels (a)–(c) are the kriged MRoS phase probabilities and panels (d)–(g) are kriged air, dewpoint and wet-bulb temperature and relative humidity. Open circles mark contributing stations, red triangles MRoS reports at that timestamp. Three features matter for the model. The fields are smooth by construction, so they represent mesoscale structure rather than local microclimate and residual sub-grid variability is absorbed into the uncertainty band rather than resolved. Predictor skill degrades away from the observation clusters, most visibly in the sparsely instrumented southern Sierra. And these are the full-data surfaces; the model predictor is the leave-one-out version, so the value a training report sees at its own location excludes that report and is necessarily noisier than shown here.

freezing the thermodynamic signal is weak and true probabilities compress toward 0.5 while away from freezing they approach 0 or 1. One map cannot serve both regimes, so we fit separate calibrators for near-freezing ($|T_w| \leq 2\,°\mathrm{C}$, consistent with the Sims and Liu 2015 uncertainty range) and clear-phase, meaning away-from-freezing, observations, routing each by its wet-bulb temperature.

Second, the choice of calibrator matters more than usual for a well-separated classifier. Isotonic regression, the conventional non-parametric choice, failed diagnostically here: held-out scores fell outside the range seen during fitting, and the out-of-range clipping mapped 62% of test predictions to exactly 0 or 1, inflating log loss while leaving the Brier score untouched. That is the signature of a calibrator reintroducing the overconfidence it was meant to remove. We therefore adopt beta calibration (Kull et al. 2017), which is parametric, monotone, and defined on the open unit interval, so it cannot produce degenerate outputs, and whose three parameters let the map be asymmetric where the score distribution is. Calibrators are fitted on final-booster scores over the pooled train and validation rain–snow observations and applied forward to the untouched test set; mixed observations are excluded entirely. Calibration adjusts probability magnitudes only, leaving rank ordering and hard decisions unchanged.

d. Handling mixed phase with an uncertainty envelope

Because rain and snow are more physically defined than mix (Section 1), standard treatments tend to mishandle that asymmetry. Discarding observer-reported mix, as most partitioning studies do, throws away the observations closest to the very transition the model is supposed to resolve. Training mix as a third class instead presumes it is a well-defined category, which it is not: mixed labels are only 11 to 12% of the total, they overlap both pure snow–rain phases across the same near-freezing temperatures rather than occupying a distinct region of feature space, and observers can apply the label inconsistently. A trained third class therefore learns mostly label noise while degrading the rain–snow boundary, which is what our first multiclass iteration found (Section c).

We take a third route: keep the supervised problem binary and treat mix as a region of irreducible uncertainty in the model's own output. Calibration settles what $p(\mathrm{snow})$ means, so the remaining question is where the binary model stops being trustworthy. A fixed threshold pair such as 0.30/0.70 is physically unjustified, because the meaning of an intermediate score depends on thermodynamic context. Near $0\,°\mathrm{C}$ wet-bulb, where snow and rain routinely coexist in the same column, $p(\mathrm{snow}) = 0.65$ is genuinely ambiguous; at $T_w = -5\,°\mathrm{C}$ the same score would instead signal a real anomaly. We therefore let the threshold width vary continuously with wet-bulb temperature through a Gaussian half-band $h(T_w)$ applied post-hoc to the calibrated probabilities:

$$\begin{aligned} h(T_w) &= b + a \exp\left(-\frac{T_w^2}{2\sigma^2}\right), \\ \text{rain threshold} &= 0.5 - h(T_w), \\ \text{snow threshold} &= 0.5 + h(T_w), \end{aligned} \tag{2}$$

The band is centered at $T_w = 0\,°\mathrm{C}$ and decays as a Gaussian of width σ, with b setting the minimum half-width far from freezing and a the additional widening at $0\,°\mathrm{C}$ (Figure 8). Any observation whose calibrated $p(\mathrm{snow})$ falls inside the band is flagged as uncertain rather than assigned a phase. We call this an *uncertainty envelope* to emphasize that it does not predict a mixed class but delineates, in probability space, the zone within which the binary model loses confident discrimination and thus abstains from making a distinct call. Observer-reported mix events then serve as an independent check on whether that zone lands where real mixed precipitation occurs.

Two metrics follow from this construction, defined in Table 2: the *flag rate*, a ground-truth independent property of model behavior, and the *mix-capture rate*, conditional on observer labels. They vary independently, since a model can flag frequently and still miss real mix if the envelope is centered wrongly, or abstain rarely and capture most of it if the envelope is well placed. Comparing the two helps explore the functionality of the model's uncertainty envelope.

The three band parameters are optimized on the validation set, which for this purpose alone includes the observed mixed events. We optimize a composite of mix capture and confident rain–snow accuracy, subject to a minimum rain–snow coverage constraint, with σ bounded to physically reasonable transition-zone widths (Sims and Liu 2015; Jennings et al. 2025). The fitted parameters are identical in both domains ($b = 0.20$, $a = 0.15$, $\sigma = 2.0\,°\mathrm{C}$), placing the thresholds at $p < 0.15$ for rain and $p > 0.85$ for snow at $0\,°\mathrm{C}$ wet-bulb and relaxing them to 0.30 and 0.70 far from freezing.

e. Evaluation framework

The framework is evaluated in three ways. A feature ablation (Section 1) retrains the model with inputs withheld to establish which sources the skill depends on. A benchmark comparison (Section 2) scores it against operational partitioning methods, overall and near freezing. And the uncertainty envelope is assessed against observer-reported mixed events (Section d). Every result is reported with a confidence interval, and every comparison as a difference with an interval of its own, obtained as described in Section 3.

(i) Metrics. A suite of complementary metrics reflects model performance more better than any one of them alone.

Each metric is sensitive to a property that the others cannot quantify: whether phase is correct on rain as well as on the dominant snow class, whether output probabilities can be trusted at face value, and whether model abstention falls on the right mix cases. Table 2 accordingly groups the metrics by what they measure. Those scoring the model's committed call establish comparability with the label-emitting benchmarks, and among these accuracy against macro F1 diagnoses class-balance effects while phase bias against accuracy separates aggregate from event-level correctness (Section d). Because a committed call discards the confidence behind it, a second group scores the probabilities directly. A third scores the flag rate, which has no counterpart in the benchmarks and cannot be reflected by accuracy; these are also the only metrics computed over mixed-phase observations, the first two groups using rain and snow alone. So that the model and the binary benchmarks are scored on identical terms, the model's committed call is taken as $p(\text{snow}) \geq 0.5$ with the envelope *not* applied. Every metric is then recomputed near freezing ($|T_w| \leq 2\,^\circ\text{C}$), where a contribution that improves the aggregate while degrading the ambiguous cases would otherwise go unnoticed.

1) Ablation design

Deciding which inputs an operational pipeline should maintain requires knowing not only which features a model uses but which it needs, and these are answered by different tools. SHapley Additive exPlanations (SHAP) attribute a trained model's predictions among its features (Lundberg et al. 2020), showing how strongly the fitted model relies on each input but not what would happen to skill if that input were unavailable. A feature ablation answers the latter directly, by retraining the identical model under controlled feature configurations. We use SHAP to identify which features the model leans on and to select which ablation configurations are worth running, then treat the retrainings as the causal evidence.

Six configurations are evaluated per domain, listed with their retained predictors in Table 4. They fall into three groups: the eight-feature baseline, leave-one-out removals isolating each assimilated source in turn, and reduced sets testing collective value (a thermodynamics-only configuration and a minimal core). All share identical splits and the same sampled predictor matrix, so differences are attributable only to the feature configuration, and all are scored on the metrics of Table 2 with near-freezing values tracked separately.

2) Benchmark methods

We compare against the standard precipitation-phase partitioning methods surveyed by Jennings et al. (2025), implemented as deterministic functions of the same gridded predictor fields: static air-temperature thresholds (T_a = 1.0 and 1.5 °C), dewpoint thresholds (T_d = 0.0 and 0.5 °C), wet-bulb thresholds (T_w = 0.0, 0.5, and 1.0 °C); the binary logistic regression of Jennings et al. (2018), $p(\text{snow}) = 1/(1 + \exp(-10.04 + 1.41\,T_a + 0.09\,\text{RH}))$; and the same logistic form re-fitted to each domain's training split, isolating the benefit of local recalibration from model form. All methods are evaluated on the identical held-out test split. Following Jennings et al. (2025), the headline comparison is binary (observer-reported mixed excluded), scored with accuracy, macro F1, and per-phase bias, reported overall and by temperature bin, with the near-freezing regime reported separately. The threshold and logistic benchmarks are strictly binary and cannot predict mixed events, so mixed-phase skill is evaluated separately from the binary comparison.

3) Confidence intervals and significance

Any metric computed on a single test set is one draw from a sampling process, and a different set of storms would have given a different value. We estimate it by bootstrapping the saved test-set prediction table (Efron and Tibshirani 1993), resampling with replacement 5,000 times and recomputing every metric on each resampled copy (a bootstrap replicate, which for readability we call a resample). Every point estimate is the metric computed once on the full test set, not a median over resamples, while the interval spans the 2.5th to 97.5th percentile of the resample values, so intervals need not be symmetric about the point and are markedly asymmetric on small subsets. All intervals are conditional on the trained model and the realized split, and quantify test-set sampling uncertainty only, not retraining variability.

Two departures from the standard procedure are needed. First, MRoS observations are not independent, because many reports duplicate one another within a storm-day, so resampling individual rows would credit the test set with more evidence than it holds and give intervals that are too narrow. We therefore resample intact clusters, defined as the observations sharing a 30-km spatial block and a calendar day, and we confirm the results are insensitive to that block size across 15, 30 and 60 km (Section d). Second, methods are compared inside each resample rather than by asking whether two separately computed intervals overlap, since model and benchmark accuracies rise and fall together from one resample to the next. Scoring all methods on identical resampled rows and recording the difference cancels that shared variability, for the same reason a paired test outperforms a two-sample test on the same data. One-sided p-values are the fraction of resamples in which the benchmark matched or exceeded the model.

5. Results

All skill values are point estimates with 95% confidence intervals, and all comparisons between methods or feature

TABLE 2. Evaluation metrics. Confusion-matrix quantities take snow as the positive class: TP and TN are correctly called snow and rain, FP is rain called snow, FN is snow called rain, and N is the number of rain and snow observations. Probability metrics use the calibrated $p(\text{snow})$, written p_i for observation i with outcome $y_i \in \{0,1\}$ (1 = snow); for ECE these are grouped into bins b with count n_b, observed snow fraction $\bar{y}_b$, and mean stated probability $\bar{p}_b$. Envelope metrics use the uncertainty band of Equation 2 and are the only ones computed over all N_{all} observations, mix included: M observer-reported mixed events, M_{in} of them inside the band, and F of all predictions inside it. Every metric is also computed on the near-freezing subset ($|T_w| \le 2\,°\text{C}$).

Metric	What it shows	Definition	Main limitation
From the confusion matrix			
Accuracy	Phases called correctly	$(\text{TP}+\text{TN})/N$	Dominated by the majority class
Snow recall	Observed snow called snow	$\text{TP}/(\text{TP}+\text{FN})$	Ignores false alarms
Rain recall	Observed rain called rain	$\text{TN}/(\text{TN}+\text{FP})$	Noisy where rain is rare
Macro F1	Precision and recall, both phases weighted equally	Mean of the two per-phase F_1, which is $2\text{TP}/(2\text{TP}+\text{FP}+\text{FN})$ for snow	Ignores the probabilities behind each call
Phase bias (%)	Over- or under-prediction of a phase	$100\,[(\text{TP}+\text{FP})/(\text{TP}+\text{FN})-1]$ for snow	Compensating errors cancel
From the predicted probabilities			
ROC AUC	Ranking: does snow score above rain?	Probability that a random snow observation scores above a random rain observation	Insensitive to calibration
Brier score	Squared error of the stated probabilities	$N^{-1}\sum_i (p_i - y_i)^2$	Mixes calibration with sharpness
ECE	Calibration: do stated probabilities verify?	$\sum_b (n_b/N)\,\lvert\bar{y}_b - \bar{p}_b\rvert$	Depends on the binning
From the uncertainty envelope			
Mix-capture rate	Observer-reported mix the model declines to call	M_{in}/M	Rises with band width
Flag rate	Predictions declined	F/N_{all}	Independent of ground truth

sets are differences computed on identical resampled data (Section 3). Intervals are shown alongside every difference, so their position does the work: an interval lying entirely above zero means the effect is larger than resampling noise can account for, while one straddling zero means the data cannot separate the effect from none, which is not the same as showing there is none. Primary performance results for both domains are collected in Table 3, per-method benchmark results in Table 5, and per-configuration ablation results in Table 4. We report overall skill first, followed by performance near freezing, ablation and benchmark results, and finally the uncertainty envelope.

a. Overall model skill

MRoS-XGB performs comparably in the two domains despite their contrasting snow climates, with a test AUROC of 0.965 in both and accuracy of 91.2% in California and 91.4% in Colorado (Table 3). Colorado's intervals are roughly three times wider throughout, which reflects its smaller test set (524 rain–snow observations against 2,301) rather than any difference in skill.

Class weighting balanced the two recalls as intended, to within 0.03 in California and 0.06 in Colorado, despite snow shares of 64% and 82% (Table 3). What remains of the imbalance appears in the phase biases, which compare how often each phase was predicted with how often it was observed. California's two bias intervals both span zero, so the model there cannot be separated from one that calls each phase exactly as often as it occurs. Colorado's do not: it calls snow 4.4% less often than observed and rain 19.8% more often. Those are the same 19 events seen from two sides, because the problem is binary and every snow observation called rain is also a rain overprediction. Nineteen events is 4.4% of Colorado's 428 test snow observations and 19.8% of its 96 rain observations, so the larger figure reflects how rare rain is there rather than careless rain calls.

Beta calibration pulled the raw scores toward the reliability diagonal in both domains (Figure A1), giving calibrated Brier scores of 0.070 and 0.058. Residual miscalibration persists at the extremes of $p(\text{snow})$, where the score distribution is strongly bimodal, and Section f traces most of it to a small, physically interpretable cluster.

b. Skill in the near-freezing regime

Accuracy falls near freezing in both domains, but far less than threshold methods lose there. On the near-freezing subset ($|T_w| \le 2\,°\text{C}$; $n = 782$ in California, $n = 179$ in Colorado), accuracy drops 5.6 and 8.7 percentage points below each domain's overall figure, while AUROC falls only from

Table 3. Held-out test-set point estimates with 95% cluster-bootstrap confidence intervals (5,000 resamples; 30-km × calendar-day clusters; percentile intervals). Point estimates are computed once on the full test set and are not medians over resamples, so intervals need not be symmetric about them. Δ rows are differences against the best benchmark in each domain (T_d = 0.0 °C in CA; T_d = 0.5 °C in CO). Intervals are conditional on the trained model and realized split.

Quantity	California	Colorado
Accuracy (%)	91.2 (89.9–92.5)	91.4 (88.4–94.1)
Near-freezing accuracy (%)	85.6 (82.9–88.3)	82.7 (76.4–88.3)
AUROC	0.965 (0.957–0.972)	0.965 (0.942–0.982)
Near-freezing AUROC	0.922 (0.903–0.941)	0.920 (0.865–0.962)
Macro F1	0.905 (0.891–0.920)	0.867 (0.821–0.906)
Near-freezing macro F1	0.855 (0.828–0.882)	0.820 (0.753–0.877)
Snow recall	0.923 (0.908–0.938)	0.925 (0.894–0.953)
Rain recall	0.893 (0.868–0.916)	0.865 (0.789–0.930)
Mix-capture rate	0.30 (0.25–0.36)	0.33 (0.21–0.45)
Δ accuracy vs. best benchmark (pts)	+11.1 (9.0–13.4)	+3.6 (0.2–6.9)
Δ near-freezing accuracy vs. best (pts)	+16.5 (12.2–21.1)	+8.4 (−0.6–16.9)

0.965 to 0.922 and 0.920 (Table 3). The model still orders snow above rain well in the regime where the model's decision to call becomes unreliable. The ablation and benchmark sections below show this is where the assimilated predictors contribute most.

Resolving skill into 1 °C wet-bulb bins shows the two domains losing accuracy in structurally different ways (Figure 4). California's snow F1 is high away from freezing, dips sharply to a minimum near T_w = 1.5 °C, and declines again above 4 °C as precipitation becomes fully liquid, while its rain F1 rises monotonically and generalizes tightly between validation and test. Colorado degrades gradually and monotonically as T_w warms, but its rain F1 is erratic above 2 °C and diverges between validation and test, which its 96 rain observations are too few to constrain. We attribute California's non-monotonic structure to its wider temperature–elevation covariation, atmospheric-river storm regimes, and bimodal elevation distribution, which together create several distinct freezing regimes where Colorado's continental climate has one. The same breakdown against air temperature (Figure A2) reproduces the structure, so it is not an artifact of the temperature variable chosen.

c. Input-dependent skill

Our first research question asks whether the assimilated predictors contribute skill beyond thermodynamics. We answer it in two steps: which features the fitted model uses, then which it needs.

(i) Feature attribution. SHAP attributions show that the two domains lean on different predictors (Figures A3 and A4). California is led by the leakage-safe MRoS surfaces and elevation, with the raw temperatures contributing surprisingly little, whereas Colorado is led by IMERG pLP and air temperature. Both shift reliance toward the phase boundary: usage rises near freezing for exactly the predictors that resolve ambiguity, among them the interpolated MRoS mix surface, even though observer-reported mix is withheld from training. That mixed observations register at all, having been discarded in prior work, is itself worth noting.

Attribution cannot settle whether a predictor is *needed*, however, because it describes only how the fitted model distributes credit among inputs that are themselves strongly correlated: the crowdsourced surfaces partly re-encode the thermodynamics they appear to displace, so their high attribution and the temperatures' low attribution are two sides of the same accounting. Retraining answers the question directly.

(ii) Ablation. Retraining under the six configurations of Table 4 confirms that the assimilated predictors carry unique information in both domains, through complementary channels. In California, removing the MRoS surfaces is the most damaging single ablation and thermodynamics alone is worse still. In Colorado no single-feature removal moves macro F1 by more than 0.01 and every interval overlaps the baseline's, so only the thermodynamics-only collapse is visible in the absolute values, and the rest we defer to the paired differences below. In both domains a compact minimal core of wet-bulb temperature, elevation, the MRoS surfaces, and IMERG pLP nearly matches the full model, indicating that a lean feature set is deployable.

The paired differences (Figure A5) turn this into causal claims and locate them near freezing. In California the MRoS surfaces are worth 9.6 points of near-freezing accuracy (6.5–12.7), and the five non-thermodynamic predictors together are worth 15.7 points (12.2–19.4), both on every metric. Three refinements emerge that the absolute values obscured. No single temperature is worth a measurable amount on overall accuracy, macro F1 or AUROC, confirming that the three substitute for one another,

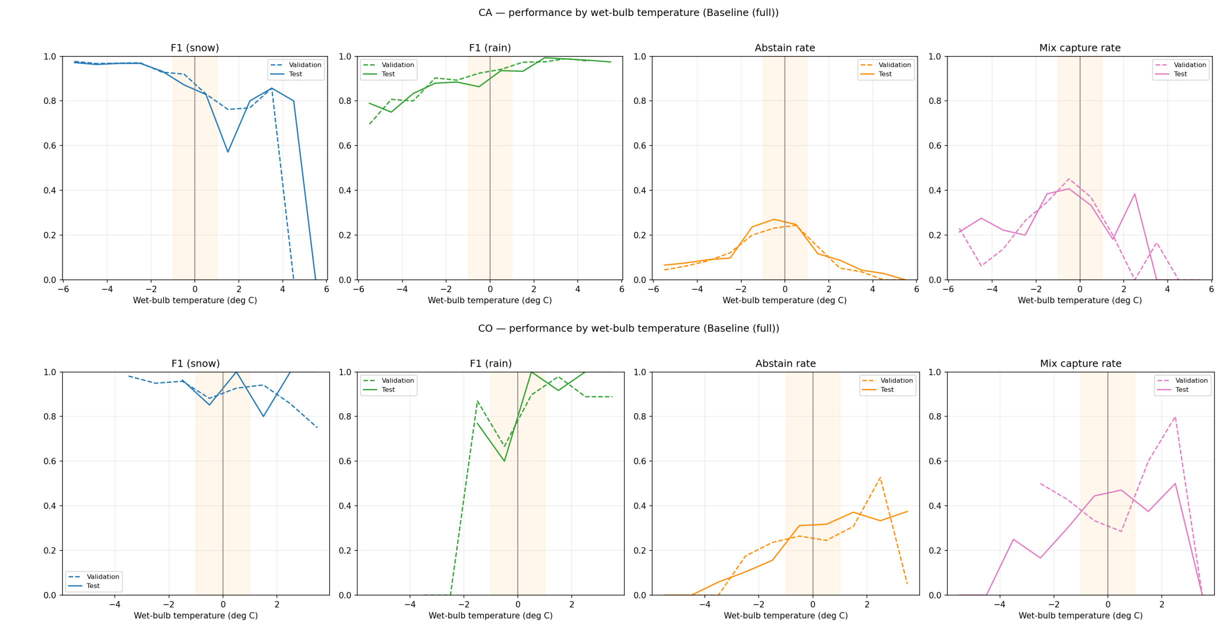


FIG. 4. Per-phase F1, flag rate, and mix-capture rate in 1 °C wet-bulb bins for California (top) and Colorado (bottom), validation (dashed) and test (solid). Shading marks $|T_w| \leq 1\,°\mathrm{C}$; bins span $-6\,°\mathrm{C}$ to $6\,°\mathrm{C}$ and are plotted at midpoints. Skill is lowest, and the uncertainty envelope most valuable, in the near-freezing transition zone. The F1 panels count only observations the model committed to, so they must be read against the flag-rate panel: a high F1 bought by frequent abstention is not the same result as a high F1 at full coverage. Line breaks mark undefined or under-sampled statistics rather than gaps in the analysis window. F1 is undefined where a bin's committed sample holds only one phase, which causes the breaks below $-2\,°\mathrm{C}$ in Colorado, and bins with fewer than 10 observations are dropped, which truncates Colorado's warm tail near 3.5 °C. A plotted zero, by contrast, is a real failure to recover a phase that was present.

with one exception: wet-bulb temperature is worth 1.7 near-freezing points (0.4–3.0), so the substitution holds in aggregate but not at the phase boundary. Elevation matters more than the absolute values suggested, contributing measurably on every metric. And IMERG pLP earns its place specifically near freezing, worth 1.9 near-freezing points (0.5–3.3) while its overall contribution stays within noise. Colorado separates fewer effects: the non-thermodynamic predictors are worth 11.2 near-freezing points there (3.1–19.5) and register on every metric, the MRoS surfaces register on overall AUROC alone, and pLP only on near-freezing AUROC, the same near-freezing role it plays in California.

Near-freezing AUROC makes the aggregate case. It rises from 0.796 (0.761–0.830) in California and 0.791 (0.715–0.862) in Colorado under thermodynamics-only to 0.922 and 0.920 for the full model, a gain of 0.126 (0.096–0.158) and 0.129 (0.066–0.196). Feature effects concentrate where the problem is hard.

Resolving the MRoS ablation against temperature (Figures 5 and A6) localizes that loss to a specific error: without the crowdsourced surfaces California's near-freezing snow recall halves, from 0.44 to 0.22 at $T_w = 1.5\,°\mathrm{C}$, so the observer network is what prevents near-freezing snow from being misread as rain. Colorado's curves cross within sampling noise, matching its weaker interval-based result.

d. Comparison against operational partitioning methods

MRoS-XGB outperformed all nine benchmarks in both domains (Table 5), and the margin widens near freezing (Figure 6; the same comparison in difference form is Figure A7). In California its accuracy was 91.2% against 80.1% for the best benchmark, a dewpoint threshold at $T_d = 0.0\,°\mathrm{C}$, and 85.6% against 69.1% near freezing. In Colorado it was 91.4% against 87.8% for $T_d = 0.5\,°\mathrm{C}$, and 82.7% against 74.3% near freezing. The benchmark ordering reproduces Jennings et al. (2025), with air-temperature-only thresholds worst and humidity-aware thresholds intermediate.

Paired differences show these gaps are not artifacts of the test draw (Figure 7). In California all 36 differences, across nine benchmarks and four metrics, are positive and clear of zero. No resample out of 5,000 favored a benchmark, giving $p < 0.0002$, which is the smallest value 5,000 resamples can resolve. Against the best benchmark MRoS-XGB gains 11.1 points of accuracy overall (9.0–13.4) and 16.5 near freezing (12.2–21.1), so even at the interval floors it is at least nine points better overall and twelve better near freezing than the strongest method in the suite.

Colorado is consistent but tighter, and there the choice of metric matters. Thirty-five of its 36 differences clear zero. On overall accuracy the gain against $T_d = 0.5\,°\mathrm{C}$ is 3.6

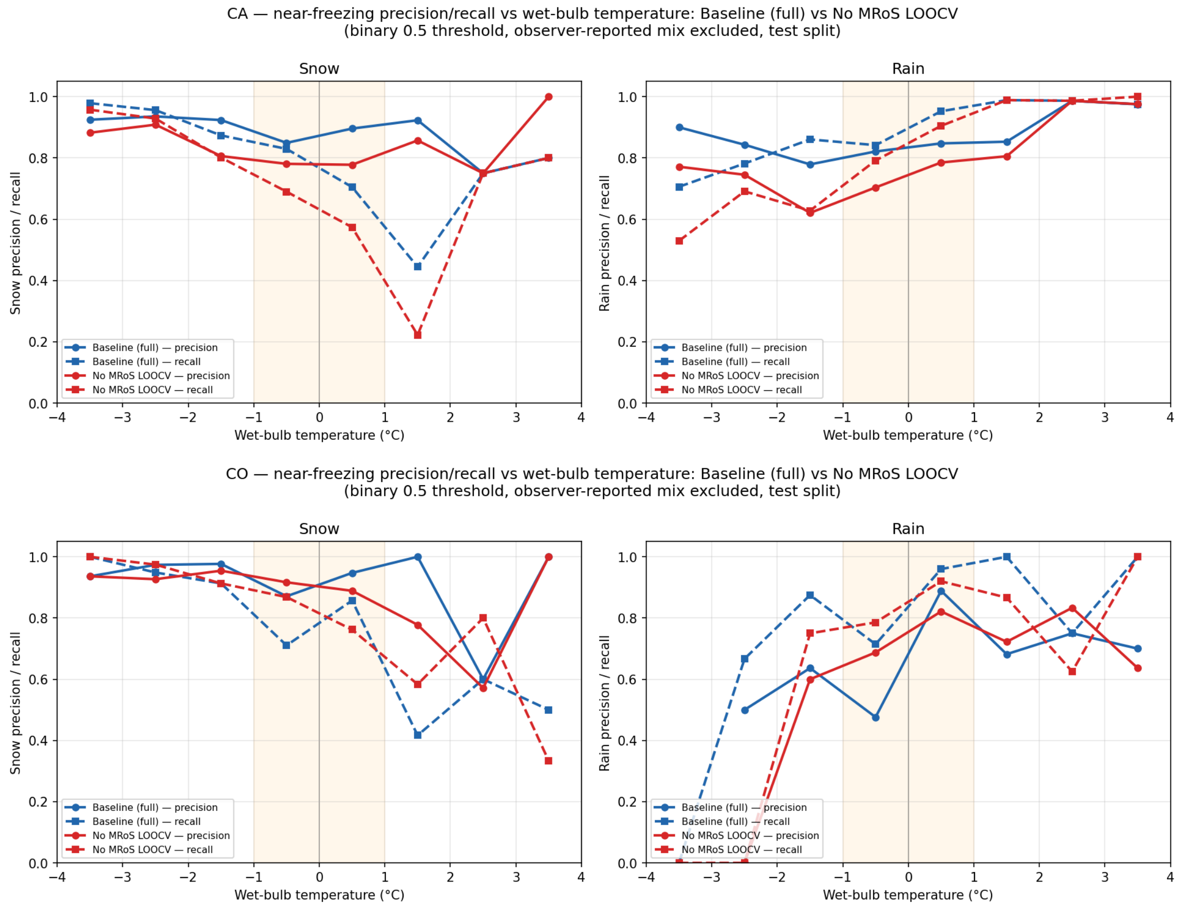


FIG. 5. What the crowdsourced surfaces contribute at the phase boundary: near-freezing precision and recall against wet-bulb temperature for the full model (blue) and the same model with the MRoS surfaces removed (red), for California (top) and Colorado (bottom), snow (left) and rain (right). Held-out test split, scored at the raw 0.5 threshold with no uncertainty band applied, so the comparison isolates the predictor contribution rather than the abstention rule; mix is excluded and shading marks $|T_w| \le 1\,°C$. In California the crowdsourced surfaces are what keeps the model usable through the transition: snow recall at $T_w = 1.5\,°C$ falls from 0.44 to 0.22 without them. Because these are recall curves at a fixed threshold, the loss appears as snow missed and called rain, which is the failure mode that matters hydrologically. Colorado does not reproduce the pattern; its curves cross repeatedly, but with 179 near-freezing observations the crossings are well inside sampling noise, and the interval-based ablation of Table 4 is the appropriate test. Figure A6 repeats the comparison against air temperature.

points (0.2–6.9, $p = 0.021$), and the near-freezing accuracy gain of 8.4 points (−0.6 to +16.9) is the one difference that does not, although 96% of resamples still favored MRoS-XGB ($p = 0.037$). On macro F1 every Colorado difference clears zero ($p \le 0.0002$), because macro F1 weights rain equally and so exposes the collapse in benchmark rain recall that accuracy conceals where 82% of observations are snow. Against $T_d = 0.5\,°C$, whose rain recall is 0.531 against MRoS-XGB's 0.865 (Table 5), the macro F1 gain is 0.096 (0.042–0.152). The domains thus tell one story at different strengths, decisive where the observer network is dense and, where it is sparse, decisive on the balanced metric while borderline on the metric the majority class dominates.

(i) Phase bias. A method can reach respectable accuracy while systematically mistaking one phase for the other, and the benchmarks do. Near freezing in California, the $T_a = 1.0\,°C$ threshold calls snow 76% less often than it occurred and rain 73% more often, while the wet-bulb thresholds fail by a similar margin in the opposite direction ($T_w = 1.0\,°C$: 74% too much snow, 71% too little rain). Because these two errors partly cancel, the accuracy figure alone conceals them, which is why we report phase bias beside it.

Only two California methods predict each phase about as often as it occurred: MRoS-XGB and the Jennings et al. (2018) logistic applied without refitting. Comparing them is instructive. The transferred logistic gets the totals right yet is correct on only 72% of individual events, so it is right on average and wrong case by case, whereas MRoS-XGB

Table 4. Ablation study: held-out test performance by feature configuration, with 95% cluster-bootstrap confidence intervals. Configurations retain the following predictors, where T_a, T_d and T_w are air, dewpoint and wet-bulb temperature, Elev is elevation, pLP is the IMERG probability of liquid precipitation, and MRoS denotes the three leakage-safe MRoS phase surfaces, counted as three predictors. *Baseline (full)*: all eight (T_a, T_d, T_w, Elev, pLP, MRoS). *No MRoS surfaces*: five (T_a, T_d, T_w, Elev, pLP). *No IMERG pLP*: seven (T_a, T_d, T_w, Elev, MRoS). *No elevation*: seven (T_a, T_d, T_w, pLP, MRoS). *Thermodynamics only*: three (T_a, T_d, T_w). *Minimal core*: six (T_w, Elev, pLP, MRoS). All configurations use kriged predictors, beta calibration, and identical splits. Single-predictor removals of T_a, T_d and T_w were also run and are discussed in Section c. NF = near-freezing ($|T_w| \leq 2\,°\mathrm{C}$).

Domain	Configuration	ROC AUC	Macro F1	NF ROC AUC	Mix capture
CA	Baseline (full)	.965 (.957–.973)	.905 (.891–.919)	.922 (.902–.941)	.30 (.24–.36)
CA	No MRoS surfaces	.936 (.925–.947)	.850 (.832–.867)	.861 (.835–.888)	.34 (.28–.40)
CA	No IMERG pLP	.964 (.956–.972)	.899 (.885–.913)	.916 (.895–.936)	.29 (.24–.35)
CA	No elevation	.956 (.947–.964)	.887 (.873–.902)	.905 (.882–.926)	.33 (.28–.38)
CA	Thermodynamics only	.907 (.892–.921)	.817 (.797–.836)	.796 (.761–.830)	.49 (.43–.55)
CA	Minimal core	.966 (.958–.973)	.900 (.885–.914)	.922 (.902–.941)	.32 (.26–.37)
CO	Baseline (full)	.965 (.941–.982)	.867 (.820–.906)	.920 (.864–.962)	.33 (.21–.45)
CO	No MRoS surfaces	.948 (.916–.973)	.863 (.820–.903)	.896 (.836–.946)	.31 (.20–.43)
CO	No IMERG pLP	.959 (.935–.978)	.871 (.825–.910)	.903 (.841–.951)	.31 (.20–.43)
CO	No elevation	.959 (.936–.977)	.862 (.818–.901)	.912 (.857–.956)	.34 (.22–.47)
CO	Thermodynamics only	.925 (.897–.951)	.798 (.746–.846)	.791 (.715–.862)	.53 (.40–.65)
CO	Minimal core	.963 (.940–.981)	.862 (.816–.903)	.924 (.873–.965)	.39 (.26–.52)

matches the totals at 91% accuracy. Predicting the right amount of snow is not the same as predicting it on the right days, and only the latter is useful for forcing a hydrologic model. Figure A8 resolves these biases against air temperature for the full benchmark suite, showing that each threshold's error reverses sign across its own threshold, so a method can appear accurate in a bin while being badly biased in both phases within it.

(ii) Sensitivity to cluster geometry. Because the cluster geometry is constructed post hoc (Section 3), we verified that it does not drive the conclusions. Intervals are essentially identical across 15-, 30-, and 60-km blocks, with California delta-CI widths of 0.042 to 0.044, and are 13 to 28% wider than an i.i.d. bootstrap would give (28% for the California headline delta, 14% for Colorado's). The clustering correction therefore mattered, but no conclusion flips between i.i.d. and clustered resampling, and none depends on the block size chosen.

e. The mixed-phase uncertainty envelope

The third research question asks whether mixed precipitation can be represented as a region of model uncertainty, which requires that the envelope fall where mixed precipitation actually occurs. It does, in the aggregate sense. The flag rate peaks near $T_w = 0\,°\mathrm{C}$ in both domains (the abstain-rate panels of Figure 4), so the model concentrates its uncertainty in the physically ambiguous regime without having been told where that regime is. Figure 8 shows the fitted envelope against the mixed events themselves, and Figure A9 resolves capture into wet-bulb bins.

Against individual events the result is partial. The envelope captured 30% (95% CI 25–36%) of observer-reported mix events in California and 33% (21–45%) in Colorado, and most misses were assigned confident extreme probabilities ($p > 0.8$ or $p < 0.2$). Captured and missed events are distributed differently in the two domains (Figure 8): in California they span a wide range of temperature and probability, suggesting meteorological ambiguity, whereas in Colorado captured events cluster near $p \approx 0.5$–0.65 and $T_w \in [-1, +2]\,°\mathrm{C}$. For context, every benchmark and the machine-learning methods of Jennings et al. (2025) capture 0%, because none incorporate mix phase or flag uncertainty. We read capture below 50% not as classifier failure but as evidence bearing on whether observer-reported mix is a coherent, learnable category (Section 6).

The envelope width is not a free choice but the optimum of an explicit tradeoff. Sweeping the three band parameters over the validation set traces a frontier between mix capture and confident rain–snow coverage. The adopted configuration ($b = 0.20$, $a = 0.15$, $\sigma = 2.0\,°\mathrm{C}$) maximizes the composite objective in California at a validation capture rate of 0.36 while retaining 90% coverage at 94.4% accuracy on the calls it makes. That figure is not comparable to the 91.2% headline accuracy, which scores every rain–snow observation including those the envelope would have flagged (Section e). Neighboring configurations trade the two monotonically: holding a and σ fixed, narrowing b to 0.15 raises California coverage to 0.92 while capture drops to 0.26, and $b = 0.10$ raises it to 0.94 as capture falls to 0.20. Colorado traces the same frontier at uniformly lower capture (0.25 at the adopted setting), consistent with its smaller mixed-event sample. Reporting the frontier rather than a single tuned point makes the operating choice auditable and lets users re-tune toward sensitivity or precision as an application demands.

TABLE 5. Held-out test performance of the model and the nine benchmark partitioning methods, with 95% cluster-bootstrap confidence intervals (5,000 resamples; 30-km × calendar-day clusters). Methods are ordered by overall accuracy within each domain. NF = near-freezing subset ($|T_w| \leq 2\,°$C; $n = 782$ in CA, $n = 179$ in CO). Point estimates are computed once on the full test set and are not medians over resamples, so intervals need not be symmetric about them. Per-phase recall and bias with intervals are given for the model in Table 3. Paired model-minus-benchmark differences, which are the basis for the significance claims in Section d, appear in Figure 7.

Domain	Method	Accuracy (%)	Macro F1	NF accuracy (%)	NF macro F1
CA	MRoS-XGB (this study)	**91.2 (89.9–92.5)**	**.905 (.891–.920)**	**85.6 (82.9–88.3)**	**.855 (.828–.882)**
CA	$T_d = 0.0\,°$C	80.1 (77.9–82.2)	.781 (.757–.802)	69.1 (65.2–72.8)	.689 (.651–.726)
CA	$T_d = 0.5\,°$C	79.1 (76.8–81.2)	.756 (.732–.779)	68.5 (64.6–72.3)	.676 (.637–.713)
CA	$T_w = 0.0\,°$C	75.5 (73.0–77.9)	.698 (.671–.725)	62.4 (58.7–66.2)	.618 (.580–.655)
CA	$T_w = 0.5\,°$C	74.8 (72.3–77.3)	.675 (.648–.702)	60.5 (56.6–64.4)	.577 (.538–.616)
CA	Bin. logistic (refit)	73.8 (71.1–76.4)	.700 (.672–.727)	60.0 (56.1–63.9)	.598 (.558–.636)
CA	$T_w = 1.0\,°$C	73.6 (70.9–76.2)	.645 (.617–.674)	56.8 (52.7–61.0)	.510 (.472–.550)
CA	Bin. logistic (transferred)	72.0 (69.3–74.6)	.697 (.670–.723)	59.5 (55.6–63.4)	.581 (.541–.620)
CA	$T_a = 1.5\,°$C	64.8 (62.0–67.5)	.645 (.618–.672)	57.0 (53.0–61.1)	.514 (.475–.552)
CA	$T_a = 1.0\,°$C	62.0 (59.2–64.8)	.620 (.592–.647)	55.1 (51.0–59.4)	.469 (.432–.508)
CO	MRoS-XGB (this study)	**91.4 (88.4–94.1)**	**.867 (.821–.906)**	**82.7 (76.4–88.3)**	**.820 (.753–.877)**
CO	$T_d = 0.5\,°$C	87.8 (84.7–90.8)	.771 (.717–.818)	74.3 (67.2–81.3)	.692 (.614–.761)
CO	$T_w = 0.5\,°$C	86.5 (82.7–89.9)	.763 (.706–.816)	70.4 (62.4–77.7)	.651 (.563–.732)
CO	$T_d = 0.0\,°$C	86.3 (82.7–89.7)	.769 (.719–.816)	71.5 (64.3–78.7)	.684 (.610–.755)
CO	$T_w = 0.0\,°$C	86.1 (82.1–89.8)	.780 (.726–.832)	69.3 (60.8–77.1)	.673 (.586–.754)
CO	$T_w = 1.0\,°$C	85.3 (81.5–88.8)	.718 (.658–.774)	67.0 (59.1–74.9)	.559 (.473–.646)
CO	Bin. logistic (refit)	84.0 (80.1–87.7)	.677 (.618–.737)	63.7 (55.8–71.4)	.514 (.439–.593)
CO	$T_a = 1.5\,°$C	81.3 (77.0–85.5)	.747 (.695–.796)	58.1 (50.3–66.1)	.581 (.500–.660)
CO	Bin. logistic (transferred)	80.7 (76.1–85.2)	.740 (.690–.790)	57.5 (49.5–65.8)	.575 (.492–.655)
CO	$T_a = 1.0\,°$C	79.2 (74.6–83.6)	.739 (.687–.788)	54.2 (45.9–62.4)	.537 (.453–.618)

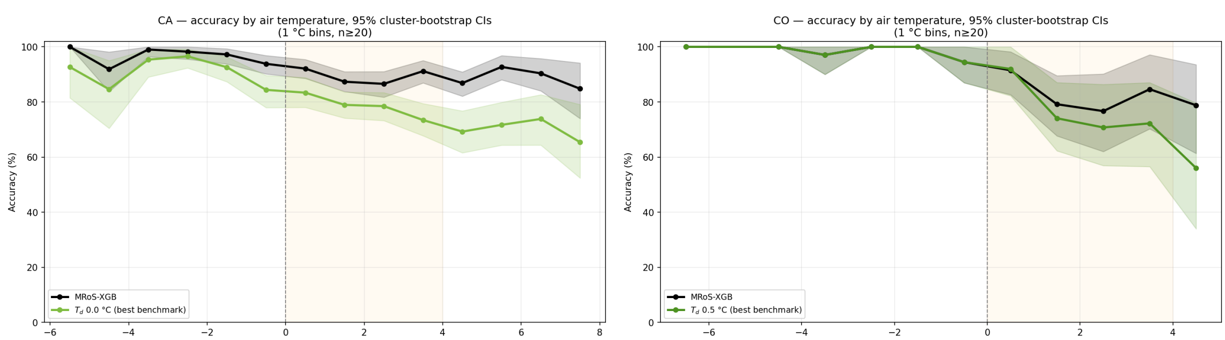


FIG. 6. Held-out accuracy by air temperature bin for MRoS-XGB (black) and the best benchmark in each domain (blue), with 95% cluster-bootstrap confidence bands, for California (left) and Colorado (right). The shaded 0 °C–4 °C region marks the climatologically ambiguous zone, where the model's advantage grows precisely as threshold methods lose skill. Accuracy is evaluated on rain and snow observations in 1 °C bins plotted at midpoints, with bins holding fewer than 20 observations dropped, which is why the curves stop short of the nominal range at both tails. Ribbons are 2.5th–97.5th percentiles of 500 cluster-bootstrap resamples over 30-km × calendar-day blocks. Resampling at cluster level makes these intervals wider than naive binomial ones, which is the honest choice given that nearby simultaneous reports are not independent; the 500 resamples used here, against 5,000 for the scalar metrics, make the tails grainier. "Best benchmark" is chosen per domain, so it is not necessarily the same method in both panels.

f. Calibration case study

Expected calibration error in California is inflated by a small, physically interpretable cluster: 42 observations, about 0.2% of the data, where calibrated $p(\text{snow}) < 0.05$ but the observer reported snow. These concentrate at lower-elevation boundary sites near 1,570 m during marginal events in which every gridded signal points toward rain, including the surrounding MRoS network via the leakage-safe surface, IMERG's retrieval weighted toward lower and warmer cells, and a near-zero wet-bulb temperature. In the Sierra Nevada the freezing level can drop rapidly over short horizontal distances, particularly during atmospheric river events, so an observer at 1,570 m may stand in snow while stations at 1,400 m record rain, and in-

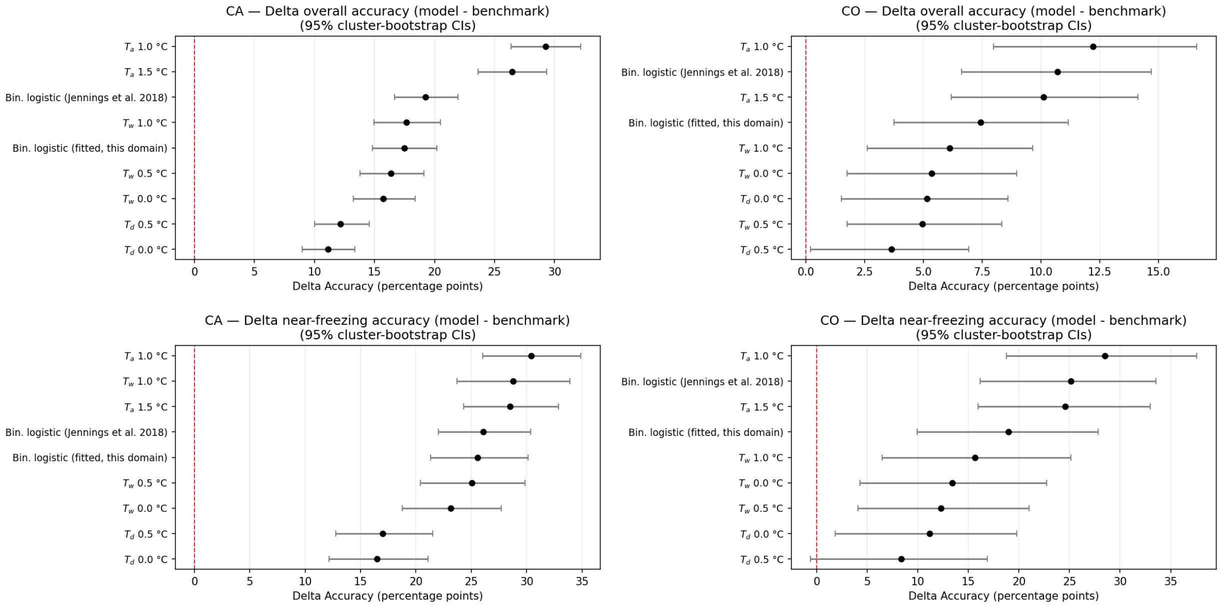


FIG. 7. Paired model-minus-benchmark accuracy differences with 95% cluster-bootstrap confidence intervals: overall (top) and near-freezing (bottom) for California (left) and Colorado (right). Every California interval lies entirely above zero. In Colorado all overall intervals do too, and only the near-freezing difference against $T_d = 0.5\,°\mathrm{C}$ reaches back across it. Differences are paired within each resample, so each interval reflects the variability of the *difference* on common data rather than of two independently resampled quantities, which is the correct construction because model and benchmark are scored on identical observations. Intervals are 2.5th–97.5th percentiles of 5,000 resamples over 30-km × calendar-day clusters; the near-freezing subset is $|T_w| \leq 2\,°\mathrm{C}$ ($n = 782$ in California, $n = 179$ in Colorado). These intervals quantify test-set sampling uncertainty only: they do not propagate uncertainty in the kriged surfaces, nor retraining variability, either of which would widen them.

terpolation across that gradient comes out rain-dominant. Within this cluster the observer is correct in roughly 75% of cases. The apparent miscalibration is therefore a property of the sparse conventional observing system rather than a modeling failure, and it is direct evidence that citizen observations carry non-redundant information exactly where conventional networks do not sample.

6. Discussion

a. Implications for phase prediction

Jennings et al. (2025) found that machine learning barely improves on threshold rules given near-surface meteorology, and our thermodynamics-only ablation reproduces that ceiling. MRoS-XGB clears it, and the ablations locate the difference in the assimilated information rather than in model form, answering the first two research questions. Both findings hold because the ceiling is set by *near-surface* information specifically: satellite pLP senses the precipitating column that no surface instrument reaches, and crowdsourced reports record the phase that actually arrived at the ground. Adding either is a new observation, not a better fit to the same predictors, and interpolation is what makes it usable, placing every sparse source on a common grid and turning the crowdsourced reports into a testable predictor surface rather than labels alone.

Where the observer network is dense the crowdsourced surfaces are load-bearing; where it is sparse, satellite pLP takes over. Substituting one source for another rather than failing outright suggests the framework should degrade gracefully as observer density declines, which is what matters for extending it to ranges without an established citizen-science network. That contribution is sharply localized: negligible away from freezing, decisive inside the transition zone, where an individual observer is often the only instrument at that elevation and moment, and is right about three times in four when contradicting every gridded input the model has (Section f). The relationship runs both ways, since models of this type can flag reports inconsistent with their surroundings while those reports reveal where gridded products are biased.

Mixed precipitation, finally, is better represented as a region of model uncertainty than as a class of its own. It has no characteristic temperature, overlaps both end members near freezing, and collects whatever observers cannot cleanly call, so training it as a third class buys mostly label noise at the cost of the rain–snow boundary, while discarding it loses the observations closest to the transition. Treating it as uncertainty instead keeps the supervised signal clean and turns the withheld mixed events into an independent test of where the model loses confidence, a test no benchmark method can take because none emits a flag

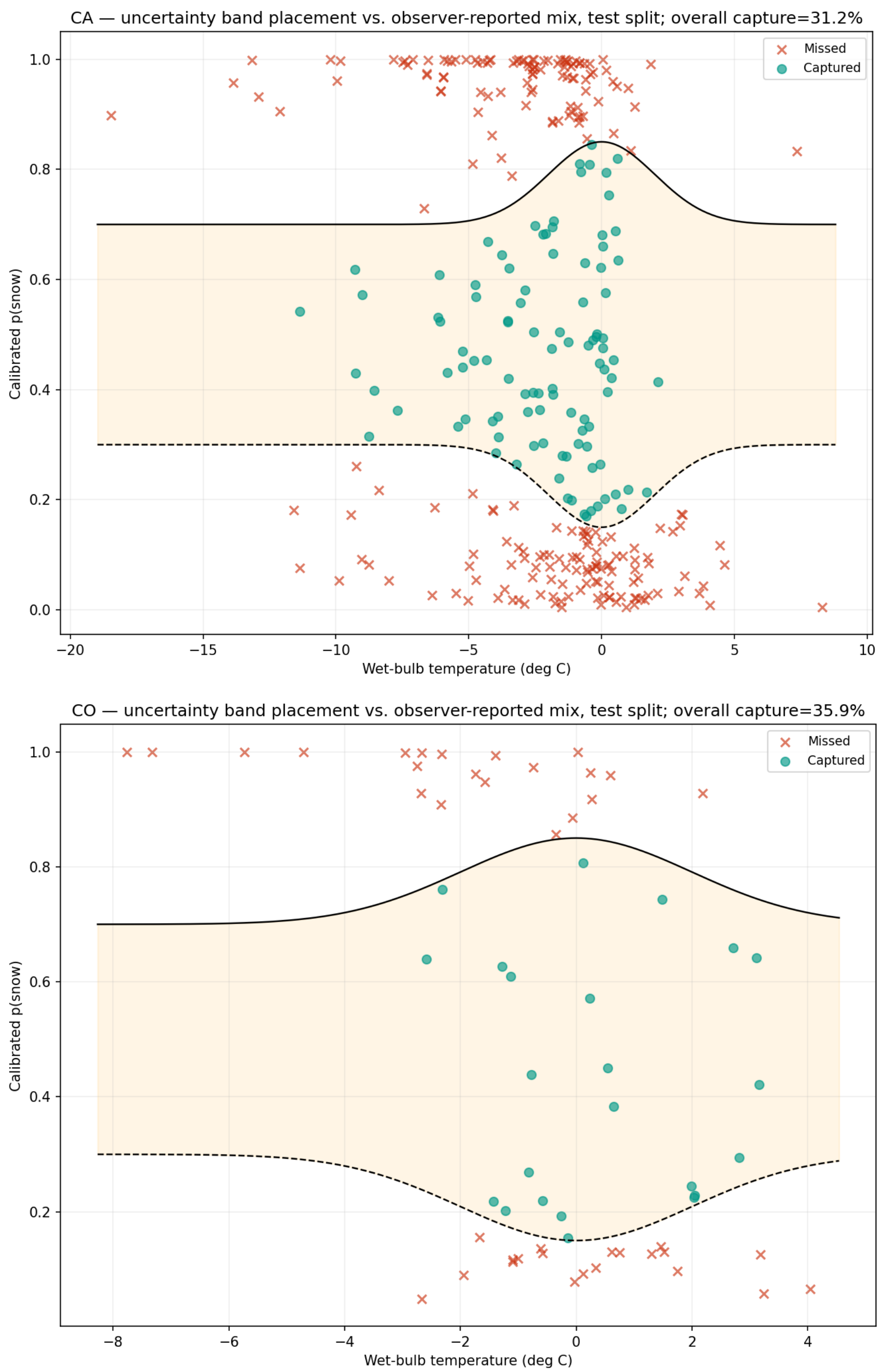


FIG. 8. The fitted uncertainty envelope and where observer-reported mixed events fall relative to it, for California (top) and Colorado (bottom), held-out test split. Solid and dashed curves are the snow and rain thresholds of Equation 2: the shaded interior is the envelope, 0.30/0.70 at both temperature extremes and opening to 0.15/0.85 at $T_w = 0°$C. Each marker is one mixed event, filled where the envelope captured it and crossed where it did not. Mix was withheld from training entirely, so every marker is an out-of-sample test. Two features carry the argument. Captured events concentrate where the envelope is widest, near 0°C, which is the behavior the construction was meant to produce. And the misses are overwhelmingly *not* near-misses: they sit against $p(\text{snow}) \approx 0$ or ≈ 1, meaning the model was confident and wrong rather than marginally short of the threshold. Widening the band would therefore recover few of them while sacrificing confident rain–snow coverage, the tradeoff quantified in Section e. Figure A9 resolves the same events by wet-bulb bin.

at all. The flag concentrates near 0 °C unprompted, though what places any single observation inside the envelope, whether genuine mixing, conflicting predictors, or sparse local data, we cannot separate.

b. Transferable methods

Two methodological results transfer beyond precipitation phase. The first concerns calibration. Gradient boosting on imbalanced classes produces a model that ranks well while its stated probabilities are unusable as probabilities, which suffices for a leaderboard but not for hydrologic forcing or for defining an interval in probability space. Isotonic regression, the default recommendation in much of the applied literature, failed diagnostically on our well-separated classifier by mapping most held-out scores to exactly 0 or 1, whereas regime-stratified beta calibration cannot. Anyone calibrating a well-separated, class-imbalanced Earth-science classifier should test for that failure rather than assume the non-parametric choice is safer.

The second concerns uncertainty in clustered observations, which describes most opportunistic and citizen-science data. Correcting for autocorrelation costs very different amounts at different stages: blocking the train–test split consumes training data a dataset of this size cannot spare, whereas resampling clusters at inference only reshapes error bars on a saved prediction table (Section 3). Where the correction is free it should be taken, and here it was not cosmetic, widening intervals by up to 28% without flipping a conclusion. Pairing matters as much: comparing methods within each resample, rather than across separately computed intervals, is what made the Colorado comparison reportable at all, and it is why we can call that result marginal rather than absent.

c. Scope and limits

The framework outputs calibrated hourly 1-km probabilities of snow and rain with an uncertainty flag, and each property has a use. The probabilities can replace static thresholds as phase forcing in hydrologic and land surface models, and because they are calibrated they can be used as probabilities rather than merely as rankings. The flag concentrates where phase is genuinely ambiguous, which is where rain-on-snow forecasting most needs human judgment. And the minimal-core result matters operationally, since six predictors nearly match eight (Table 4) and each one dropped is an upstream dependency removed, IMERG latency being the main deployment risk.

The claims have boundaries. The randomized split may admit mild within-storm leakage, so point estimates may be slightly optimistic for a novel season; the cluster bootstrap widens intervals but cannot repair a point estimate, and all intervals are conditional on the trained model and split. The MRoS network is opportunistic and skewed toward higher elevations, roads, and daylight hours, and observer definitions are inconsistent, particularly for mix. As a gridded product the framework inherits point-to-grid representativeness error, because a 1-km cell average cannot capture sub-grid phase transitions. And the results rest on two domains and one observation network, so transferability remains to be demonstrated.

The most promising extension follows from the ceiling argument itself. Skill near freezing remains the weakest regime regardless of feature set, because satellite pLP resolves the precipitating column only coarsely. Predictors describing that column directly, such as profile temperatures and freezing level, are the natural next addition, and the framework is built to accept them.

7. Conclusions

We developed a framework that assimilates citizen-science phase observations, station meteorology, satellite pLP, and terrain onto hourly 1-km grids and trains a calibrated binary XGBoost classifier with an uncertainty-derived mixed-phase flag. Across two contrasting mountain domains MRoS-XGB achieved a test AUROC of 0.965 in each and outperformed all standard partitioning benchmarks, with the largest gains near freezing, where phase uncertainty is most consequential (85.6% accuracy against 69.1% for the best California benchmark). Ablations trace those gains to the assimilated information rather than to model form, which is consistent with a skill ceiling set by near-surface meteorology, and the calibrated probabilities make the outputs usable as probabilities rather than rankings. The uncertainty envelope localizes low-confidence predictions in the physically ambiguous near-freezing zone, offering an operational alternative to forcing confident calls on genuinely ambiguous events. A paired cluster bootstrap places these conclusions on firm footing: every California model-minus-benchmark difference is positive beyond resampling noise ($p < 0.0002$), with the advantage over the best benchmark at least nine points overall and twelve near freezing at the interval floor, while Colorado's smaller sample yields consistent but borderline significance on accuracy and unambiguous gains on macro F1. The framework is decisive where the observer network is dense and directionally consistent where it is sparse.

Acknowledgments. We thank the Mountain Rain or Snow volunteer observers, whose contributions make this work possible. **[TODO: funding sources and grant numbers (required by AMS), program officers, computing resources, and a conflict-of-interest statement.]**

Data availability statement. **[TODO: State where the data and code underlying the findings can be accessed, with persistent DOIs: MRoS observation archive (DOI); station data (HADS/LCD/WCC access points); GPM IMERG (GES DISC DOI); PRISM; USGS 3DEP. Code: pipeline repository (GitHub link**

plus archived Zenodo DOI), including the ten-stage workflow scripts and ML notebooks, with license.]

APPENDIX

Supplementary Figures

This appendix collects supporting figures, in the order they are first cited. Each resolves a main-text result into more detail or repeats it against a second binning variable as a robustness check, and none is required to follow the argument. Figure A1 gives reliability diagrams and calibrated score distributions. Figure A2 resolves skill against air temperature, the counterpart to Figure 4. Figures A3 and A4 give the SHAP attribution summarized in Section c, the first grouped by predicted phase and the second resolved into wet-bulb bins. Figure A5 presents the ablation differences with intervals, and Figure A6 repeats Figure 5 against air temperature. Figure A8 shows accuracy and per-phase bias against air temperature for all nine benchmarks, Figure A7 states the model's advantage in paired difference form, and Figure A9 resolves mix capture into wet-bulb bins.

References

Arienzo, M. M., M. Collins, and K. S. Jennings, 2021: Enhancing engagement of citizen scientists to monitor precipitation phase. *Frontiers in Earth Science*, **9**, 617 594, https://doi.org/10.3389/feart.2021.617594.

Barnett, T. P., J. C. Adam, and D. P. Lettenmaier, 2005: Potential impacts of a warming climate on water availability in snow-dominated regions. *Nature*, **438 (7066)**, 303–309, https://doi.org/10.1038/nature04141.

Chen, T., and C. Guestrin, 2016: XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794, https://doi.org/10.1145/2939672.2939785.

Cressie, N. A. C., 1993: *Statistics for Spatial Data*. Revised ed., John Wiley & Sons, New York, https://doi.org/10.1002/9781119115151.

Dai, A., 2008: Temperature and pressure dependence of the rain–snow phase transition over land and ocean. *Geophysical Research Letters*, **35 (12)**, L12 802, https://doi.org/10.1029/2008GL033295.

Daly, C., M. Halbleib, J. I. Smith, W. P. Gibson, M. K. Doggett, G. H. Taylor, J. Curtis, and P. P. Pasteris, 2008: Physiographically sensitive mapping of climatological temperature and precipitation across the conterminous United States. *International Journal of Climatology*, **28 (15)**, 2031–2064, https://doi.org/10.1002/joc.1688.

Ding, B., K. Yang, J. Qin, L. Wang, Y. Chen, and X. He, 2014: The dependence of precipitation types on surface elevation and meteorological conditions and its parameterization. *Journal of Hydrology*, **513**, 154–163, https://doi.org/10.1016/j.jhydrol.2014.03.038.

Efron, B., and R. J. Tibshirani, 1993: *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 436 pp. pp., https://doi.org/10.1201/9780429246593.

Feiccabrino, J., W. Graff, A. Lundberg, N. Sandström, and D. Gustafsson, 2015: Meteorological knowledge useful for the improvement of snow–rain separation in surface-based models. *Hydrology*, **2 (4)**, 266–288, https://doi.org/10.3390/hydrology2040266.

Goovaerts, P., 1997: *Geostatistics for Natural Resources Evaluation*. Oxford University Press, New York.

Harder, P., and J. Pomeroy, 2013: Estimating precipitation phase using a psychrometric energy balance method. *Hydrological Processes*, **27 (13)**, 1901–1914, https://doi.org/10.1002/hyp.9799.

Harpold, A. A., M. L. Kaplan, P. Z. Klos, T. Link, J. P. McNamara, S. Rajagopal, R. Schumer, and C. M. Steele, 2017: Rain or snow: hydrologic processes, observations, prediction, and research needs. *Hydrology and Earth System Sciences*, **21 (1)**, 1–22, https://doi.org/10.5194/hess-21-1-2017.

Hengl, T., G. B. M. Heuvelink, and D. G. Rossiter, 2007: About regression-kriging: From equations to case studies. *Computers & Geosciences*, **33 (10)**, 1301–1315, https://doi.org/10.1016/j.cageo.2007.05.001.

Huffman, G. J., and Coauthors, 2020: Integrated multi-satellite retrievals for GPM (IMERG) technical documentation. Tech. rep., NASA Goddard Space Flight Center. Or cite Huffman et al. (2020) IMERG chapter in Satellite Precipitation Measurement, Springer; verify preferred citation.

Jennings, K. S., M. M. Arienzo, M. Collins, B. J. Hatchett, A. W. Nolin, and G. Aggett, 2023: Crowdsourced data highlight precipitation phase partitioning variability in rain–snow transition zone. *Earth and Space Science*, **10 (3)**, e2022EA002 714, https://doi.org/10.1029/2022EA002714.

Jennings, K. S., and N. P. Molotch, 2019: The sensitivity of modeled snow accumulation and melt to precipitation phase methods across a climatic gradient. *Hydrology and Earth System Sciences*, **23 (9)**, 3765–3786, https://doi.org/10.5194/hess-23-3765-2019.

Jennings, K. S., T. S. Winchell, B. Livneh, and N. P. Molotch, 2018: Spatial variation of the rain–snow temperature threshold across the Northern Hemisphere. *Nature Communications*, **9**, 1148, https://doi.org/10.1038/s41467-018-03629-7.

Jennings, K. S., and Coauthors, 2025: Machine learning shows a limit to rain–snow partitioning accuracy when using near-surface meteorology. *Nature Communications*, **16**, https://doi.org/10.1038/s41467-025-58234-2.

Journel, A. G., 1983: Nonparametric estimation of spatial distributions. *Journal of the International Association for Mathematical Geology*, **15 (3)**, 445–468, https://doi.org/10.1007/BF01031292.

Kull, M., T. M. Silva Filho, and P. Flach, 2017: Beta calibration: a well-founded and easily implemented improvement on logistic calibration for binary classifiers. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, Vol. 54, 623–631.

Li, D., M. L. Wrzesien, M. Durand, J. Adam, and D. P. Lettenmaier, 2017: How much runoff originates as snow in the western United States, and how will that change in the future? *Geophysical Research Letters*, **44 (12)**, 6163–6172, https://doi.org/10.1002/2017GL073551.

Lundberg, S. M., and Coauthors, 2020: From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, **2 (1)**, 56–67, https://doi.org/10.1038/s42256-019-0138-9.

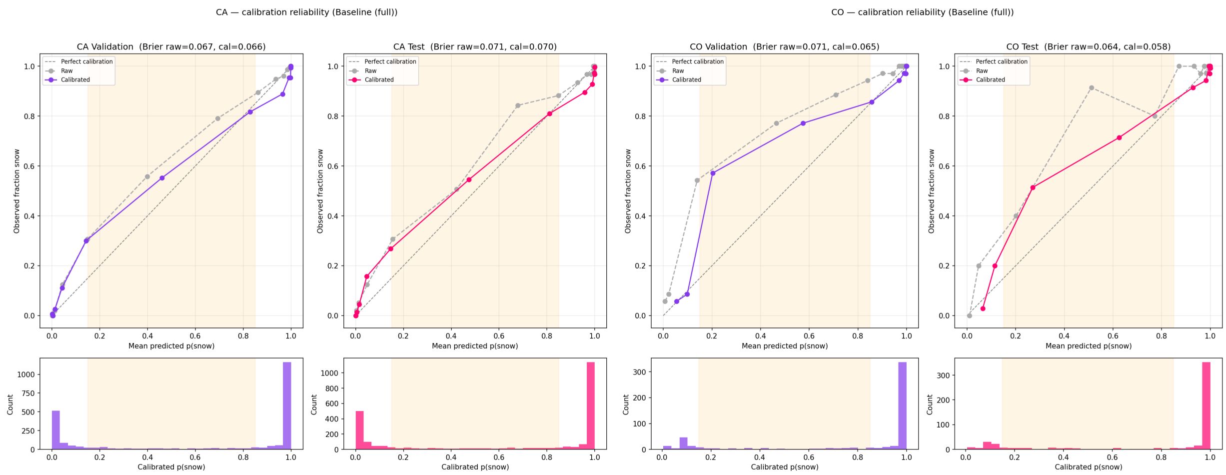


Fig. A1. Reliability diagrams and calibrated score distributions for California (left pair) and Colorado (right pair), validation and test. Beta calibration (colored) moves the raw XGBoost curve (gray) toward the diagonal; the shaded region marks the maximum extent of the uncertainty band at $T_w = 0\,^\circ$C (0.15–0.85). The bimodal score histograms show that most predictions are confident, with the band occupying the sparse middle. Reliability curves use 15 quantile-spaced bins, so each point carries roughly equal sample weight and the visible spacing of points along the abscissa is itself informative: the wide gap across the middle of the range reflects how few predictions land there. Only pure snow and rain observations enter these diagrams, since calibration is defined against a binary outcome and observer-reported mix has no corresponding target probability; mix events are therefore absent here and assessed separately in Figure 8. Brier scores in the panel titles are computed on the same pure-phase subset, raw versus calibrated. Residual departures from the diagonal at the extreme bins should be read with the histogram beneath them, where the small denominators mean a handful of observations can move a bin's observed frequency substantially.

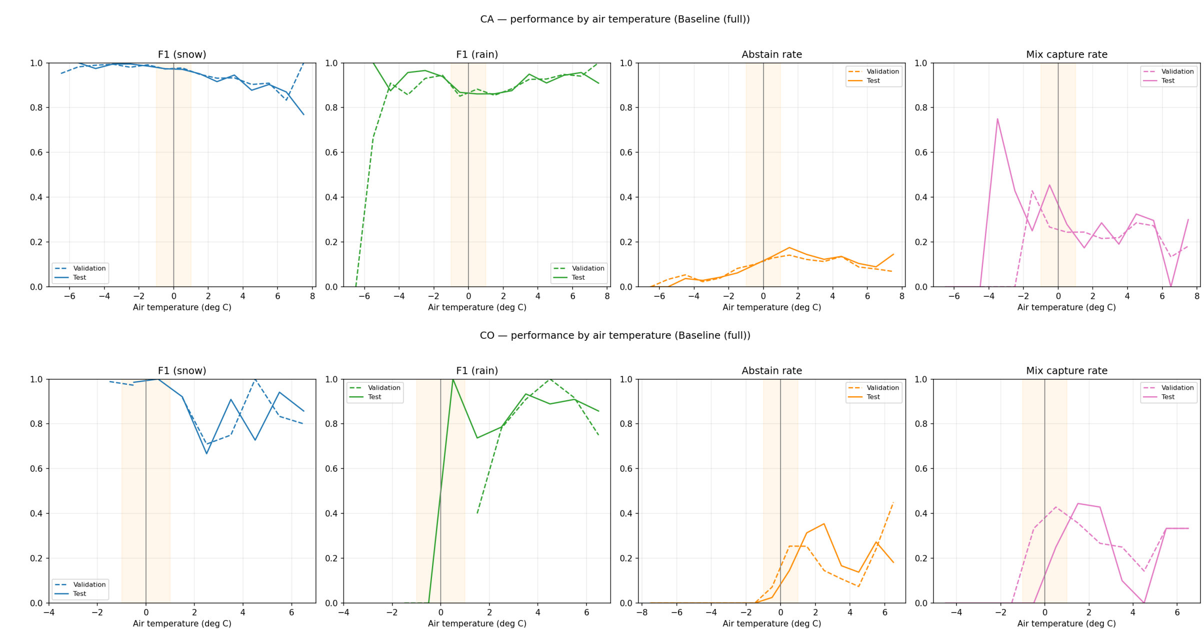
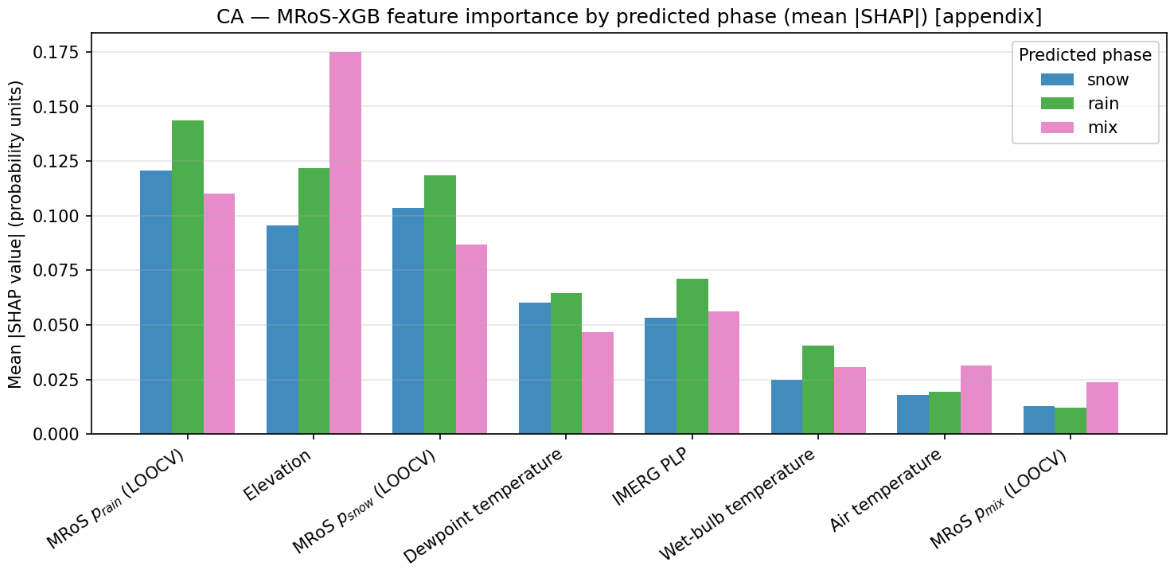


Fig. A2. Model performance in 1 °C air-temperature bins for California (top) and Colorado (bottom), validation (dashed) and test (solid), with shading marking ±1 °C about freezing. Panels show snow F1, rain F1, the flag rate, and the mix-capture rate. Two points follow. This is the air-temperature counterpart to the wet-bulb view of Figure 4 and reproduces the same structure, with snow F1 declining as temperature rises, rain F1 improving, and the flag rate peaking just above 0 °C, which confirms the envelope concentrates where phase is ambiguous under either temperature variable. And Colorado's curves are visibly noisier throughout, reflecting its smaller sample. Conventions follow Figure 4, except that bins span −8 °C to 8 °C rather than −6 °C to 6 °C, since air and wet-bulb temperature are physically distinct and reusing one set of edges would not be meaningful. F1 is computed only over committed observations, bins with fewer than 10 observations are dropped, and F1 is undefined where a bin holds a single phase, which is what interrupts Colorado's curves below −2 °C. A plotted zero, as in Colorado's test rain F1 near −0.5 °C, instead means both phases were present and the model recovered no rain.



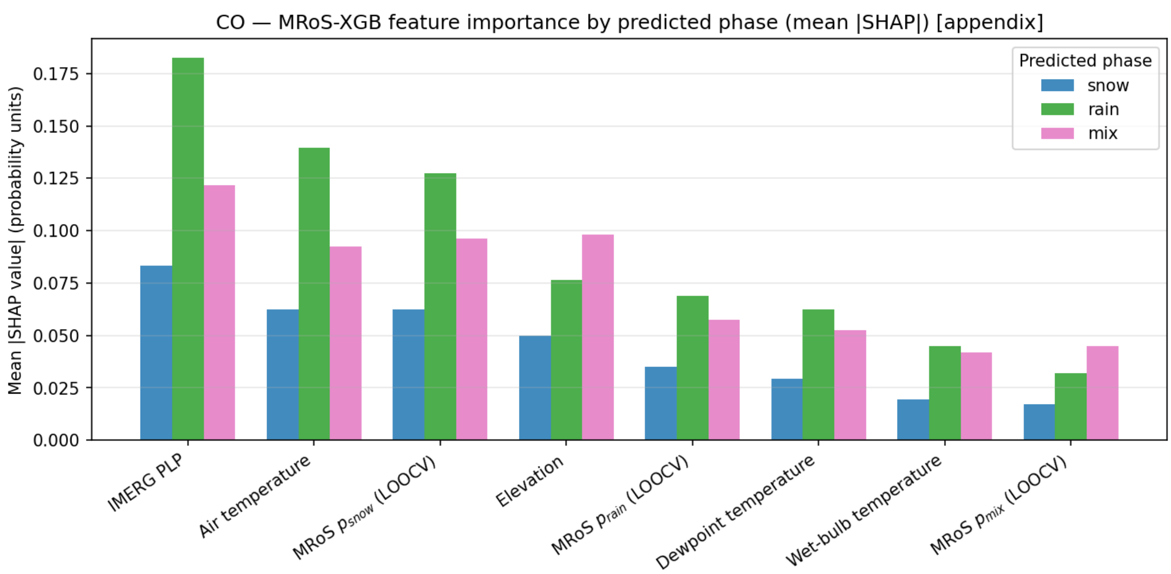


FIG. A3. Mean absolute SHAP value per predictor, grouped by the phase the model predicted, for California (top) and Colorado (bottom). Bars are ordered by overall importance, in probability units, so a bar of 0.10 means the predictor moves $p\,(\text{snow})$ by 0.10 on average. The domains rank their predictors differently: California leads with the crowdsourced surfaces and elevation, Colorado with IMERG pLP and air temperature. Two features carry interpretive weight. Attributions are larger for rain and mix predictions than for snow in both domains, indicating the model needs more evidence to move away from the snow-dominant base rate than to stay with it. And correlation among predictors shapes these magnitudes: in California the MRoS snow and rain surfaces are anticorrelated at $r = -0.92$, air and wet-bulb temperature at 0.81, and the MRoS snow surface with IMERG pLP at -0.75, so the crowdsourced surfaces are a spatial re-encoding of the thermodynamics rather than an independent signal, which is why credit transfers onto them and away from the raw temperatures. Three limits apply to reading the bars. Groups are by *predicted* phase, so "mix" is the set of predictions the band flagged rather than observed mix events. Mean absolute SHAP discards sign, so a large bar means strong response, not a push toward snow. And attributions are computed against each domain's own baseline, so bars compare within a panel but only loosely across domains.

Lundquist, J. D., P. J. Neiman, B. Martner, A. B. White, D. J. Gottas, and F. M. Ralph, 2008: Rain versus snow in the Sierra Nevada, California: Comparing doppler profiling radar and surface observations of melting level. *Journal of Hydrometeorology*, **9 (2)**, 194–211, https://doi.org/10.1175/2007JHM853.1.

Minder, J. R., D. R. Durran, and G. H. Roe, 2011: Mesoscale controls on the mountainside snow line. *Journal of the Atmospheric Sciences*, **68 (9)**, 2107–2127, https://doi.org/10.1175/JAS-D-10-05006.1.

Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno), 2024: rainOrSnowTools: An R package

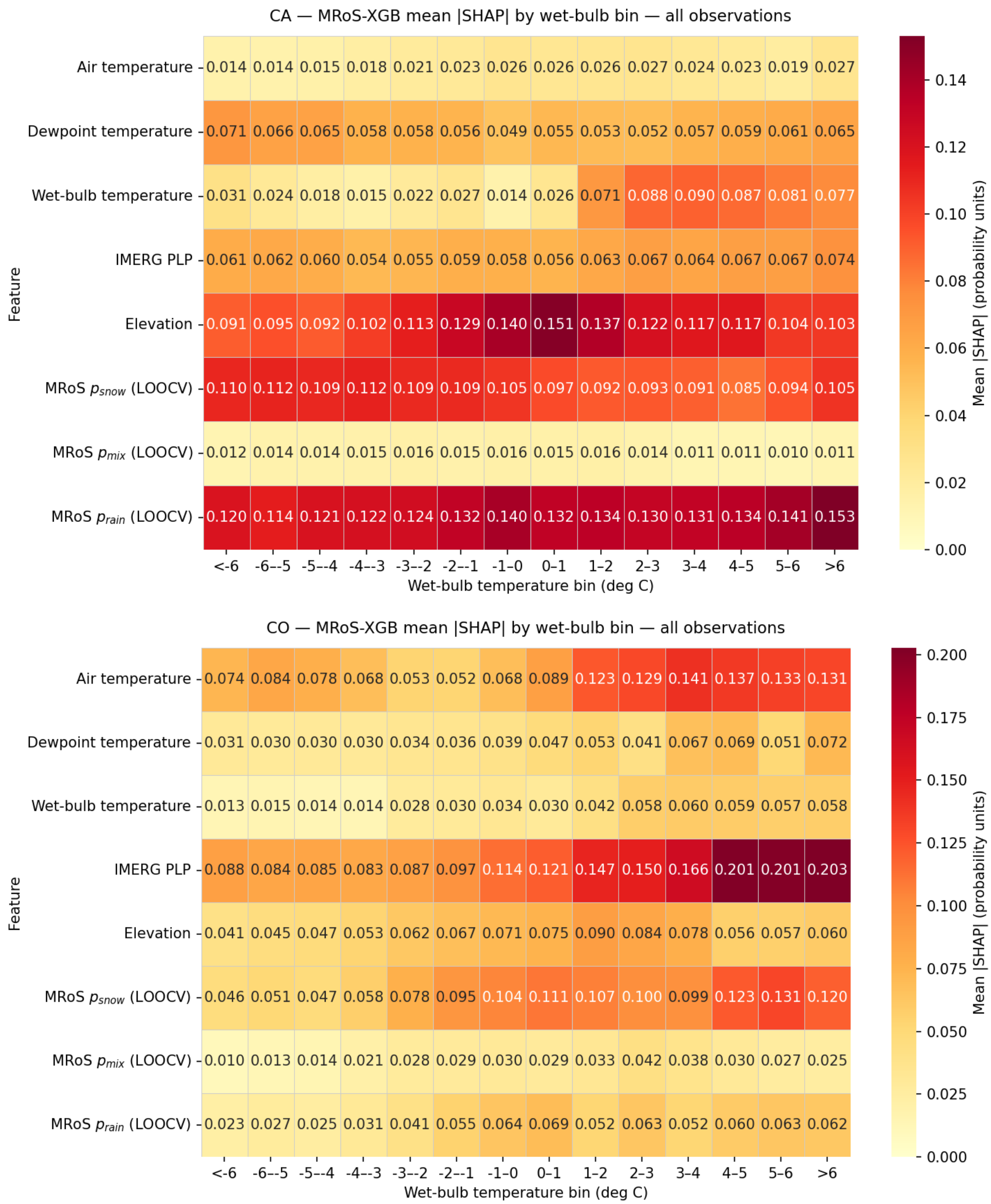


Fig. A4. Mean absolute SHAP value per predictor within 1 °C wet-bulb bins, over all observations, for California (top) and Colorado (bottom). Whereas Figure A3 gives one number per predictor, this view resolves how the model reallocates reliance across the thermodynamic range. Two contrasts matter. California's reliance on elevation peaks sharply in the transition zone (0.151 at T_w = 0 to 1 °C) while wet-bulb temperature contributes little below 1 °C and rises several-fold once phase is unambiguous, whereas Colorado's reliance on IMERG pLP climbs monotonically from 0.088 in the coldest bin to 0.203 in the warmest. That is the attribution-side counterpart of the ablation result: California recruits terrain and the crowdsourced surfaces where the thermodynamic signal is weakest, while Colorado leans on the satellite column signal throughout. Summarized as a ratio of mean |SHAP| near freezing to away from freezing, usage rises for air temperature (1.44×), elevation (1.38×), the MRoS mix surface (1.14×) and rain surface (1.10×), and falls for dewpoint (0.84×). Bin populations are strongly unequal, so a bright cell in an outer bin rests on far fewer observations than one near freezing, and empty bins are omitted rather than shown as zero. As in Figure A3 the quantity is unsigned, so these rows show where the model draws on a predictor, not which way it pushes the score.

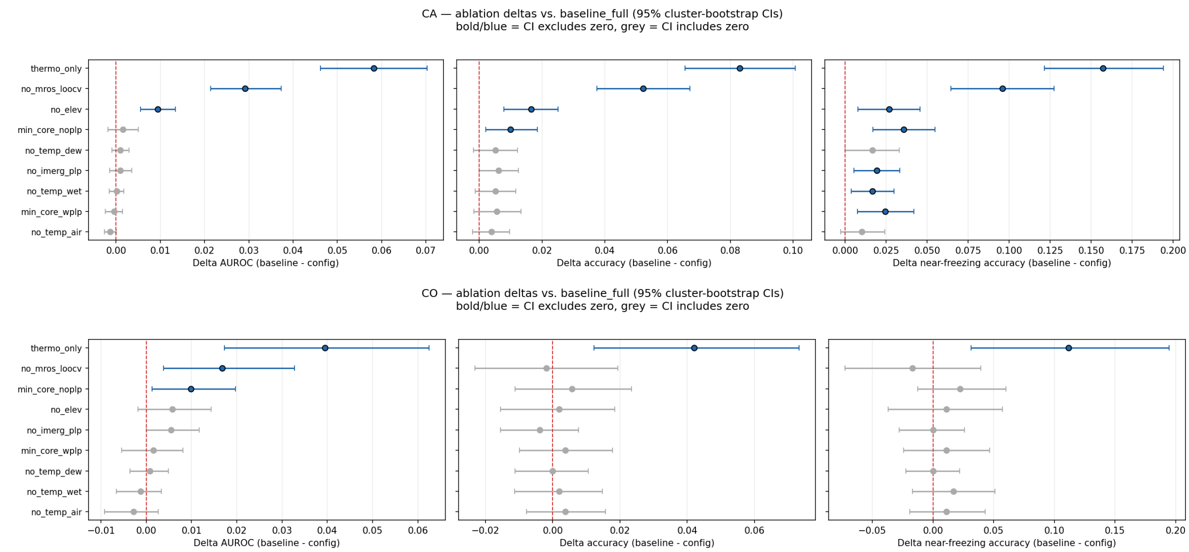


Fig. A5. Ablation differences from the full model with 95% cluster-bootstrap confidence intervals, for California (top) and Colorado (bottom): AUROC, overall accuracy, and near-freezing accuracy, each as baseline minus configuration, so positive values mean the removed predictors were contributing. Intervals are 2.5th–97.5th percentiles of 5,000 resamples over 30-km × calendar-day clusters, paired within resample; markers are drawn in blue with a dark edge where the interval lies clear of zero and grey where it crosses zero, and configurations are ordered by the difference in the leftmost panel so vertical position is comparable across panels. This is the interval-based counterpart to the point estimates underlying Table 4, and it is the version that supports the claims in Section c: the thermodynamics-only configuration separates cleanly from the full model on every metric in both domains, whereas most single-predictor removals stay within noise, which is the signature of predictor redundancy rather than of predictors without value. Near-freezing accuracy carries wider intervals than overall accuracy throughout, since it is computed on the $|T_w| \leq 2\,°\mathrm{C}$ subset alone ($n = 782$ in California, $n = 179$ in Colorado); differences that appear only there and not in the overall metric should be read as suggestive. No macro-F1 difference was archived in the bootstrap output, so that metric appears in Table 4 but not here.

for processing Mountain Rain or Snow citizen science observations. Software; verify authors, version, and release DOI before submission, https://github.com/LynkerIntel/rainOrSnowTools.

Muller, C. L., L. Chapman, S. Johnston, C. Kidd, S. Illingworth, G. Foody, A. Overeem, and R. R. Leigh, 2015: Crowdsourcing for climate and atmospheric sciences: current status and future potential. *International Journal of Climatology*, **35 (11)**, 3185–3203, https://doi.org/10.1002/joc.4210.

Musselman, K. N., F. Lehner, K. Ikeda, M. P. Clark, A. F. Prein, C. Liu, M. Barlage, and R. Rasmussen, 2018: Projected increases and shifts in rain-on-snow flood risk over western North America. *Nature Climate Change*, **8 (9)**, 808–812, https://doi.org/10.1038/s41558-018-0236-4.

Sims, E. M., and G. Liu, 2015: A parameterization of the probability of snow–rain transition. *Journal of Hydrometeorology*, **16 (4)**, 1466–1477, https://doi.org/10.1175/JHM-D-14-0211.1.

Stull, R., 2011: Wet-bulb temperature from relative humidity and air temperature. *Journal of Applied Meteorology and Climatology*, **50 (11)**, 2267–2269, https://doi.org/10.1175/JAMC-D-11-0143.1.

Wackernagel, H., 2003: *Multivariate Geostatistics: An Introduction with Applications*. 3rd ed., Springer, Berlin, https://doi.org/10.1007/978-3-662-05294-5.

Wang, Y.-H., P. Broxton, Y. Fang, A. Behrangi, M. Barlage, X. Zeng, and G.-Y. Niu, 2019: A wet-bulb temperature-based rain–snow partitioning scheme improves snowpack prediction over the drier western United States. *Geophysical Research Letters*, **46 (23)**, 13 825–13 835, https://doi.org/10.1029/2019GL085722.

Wayand, N. E., J. Stimberis, J. P. Zagrodnik, C. F. Mass, and J. D. Lundquist, 2016: Improving simulations of precipitation phase and snowpack at a site subject to cold air intrusions: Snoqualmie Pass, WA. *Journal of Geophysical Research: Atmospheres*, **121 (17)**, 9929–9942, https://doi.org/10.1002/2016JD025387.

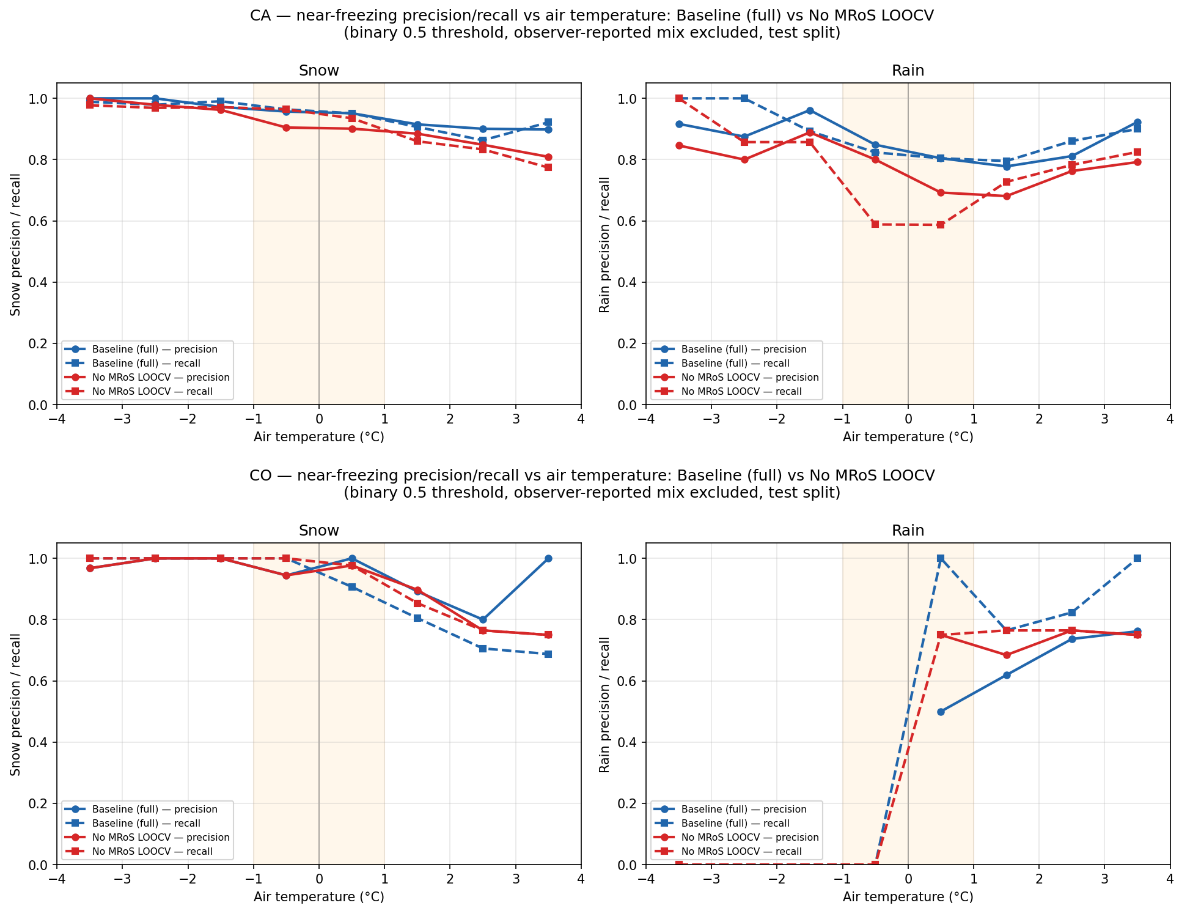


Fig. A6. Air-temperature counterpart to Figure 5: near-freezing precision and recall for the full model versus the model without the MRoS LOOCV surfaces, in 1 °C air-temperature bins on the held-out test split, scored at the raw 0.5 threshold with mix excluded. Binning against air temperature relocates the effect rather than removing it. In California the loss now appears in rain recall instead of snow recall — 0.82 falling to 0.59 at −0.5 °C and 0.80 to 0.59 at 0.5 °C — because air temperature discriminates phase less sharply than wet-bulb temperature, so the bins mix thermodynamic regimes that the wet-bulb view separates. That the attributable loss survives the change of binning variable, while moving between phases, is the robustness point: the crowdsourced surfaces are supplying information the temperature fields do not carry, not sharpening a particular temperature threshold. Colorado's rain curves are undefined or near-degenerate in several cold bins here, where its rain sample is only a handful of observations, and should not be interpreted.

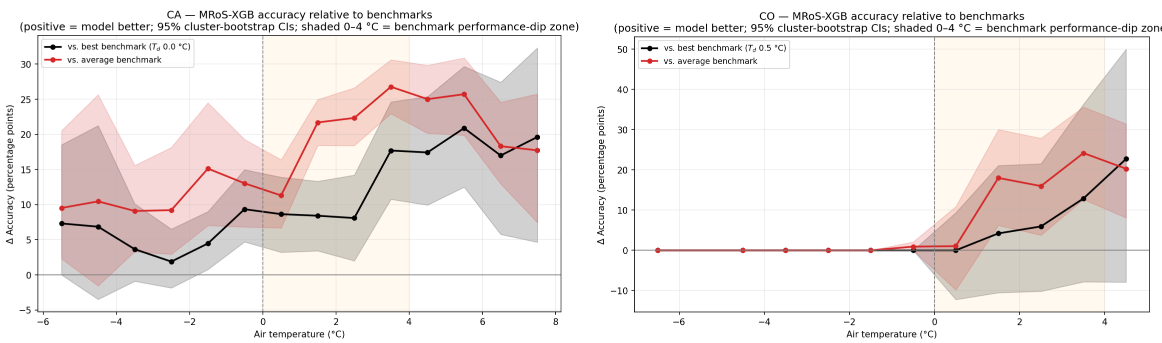


Fig. A7. Model accuracy relative to the benchmarks by air-temperature bin, California (left) and Colorado (right): percentage-point difference against the best single benchmark (black) and against the mean of all nine (red), with 95% cluster-bootstrap ribbons, positive meaning the model is more accurate. This is the difference form of Figure 6, and it states the paper's central benchmark claim compactly: the advantage is smallest in the cold bins, where every method gets phase right, and rises into the shaded 0 °C–4 °C zone where thresholds fail, reaching roughly 18 percentage points against the best benchmark and 27 against the benchmark average in California. Two cautions. Differences are paired within each resample, so the ribbons describe uncertainty in the difference rather than in either accuracy separately, and the vs. average-benchmark line is not a comparison against any deployable method — it summarizes the family and is the more favorable of the two framings. The cold-bin ribbons cross zero in both domains, so we do not claim an advantage there; the claim is specifically about the transition zone.

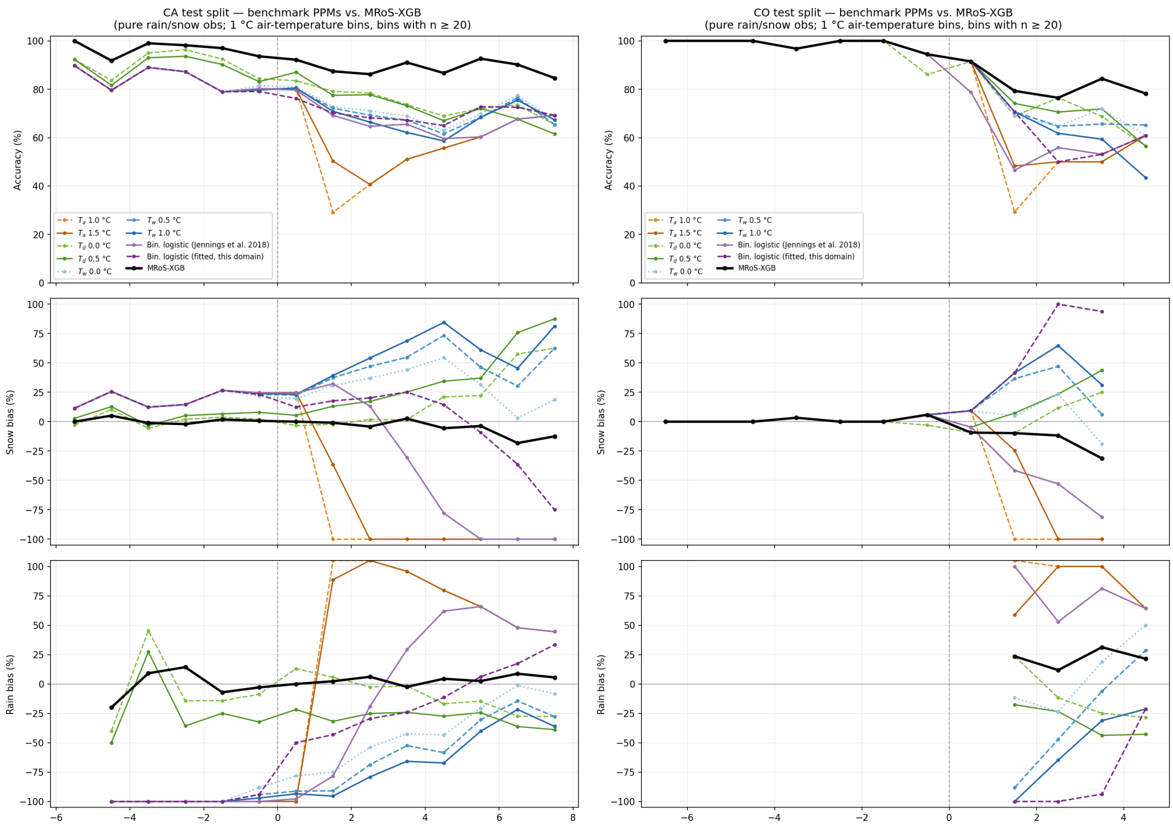


FIG. A8. Accuracy and per-phase bias against air temperature for all nine benchmark partitioning methods and the model, California (left) and Colorado (right), held-out test split, 1 °C bins with $n \geq 20$ pure observations. Panels give, from top to bottom, accuracy, snow bias, and rain bias, where bias is the signed difference between predicted and observed frequency of that phase within the bin, so zero is unbiased and positive means over-prediction. These are point estimates without intervals; Figure 6 gives the bootstrap-resampled version for the model and best benchmark. The bias panels are the reason to include this figure: they show *how* fixed-threshold methods fail rather than only that they do. Each threshold method's error is a compensating pair — systematic snow over-prediction on the warm side of its threshold and rain over-prediction on the cold side — so its accuracy in any single bin understates the structure of the error, and bins where two methods tie on accuracy can be biased in opposite directions. The model's bias curves stay closer to zero across the range rather than trading one phase against the other, which is what a threshold-free partition should look like. Colder and warmer bins are dropped where fewer than 20 pure observations are available, so the curves do not all span the same range.

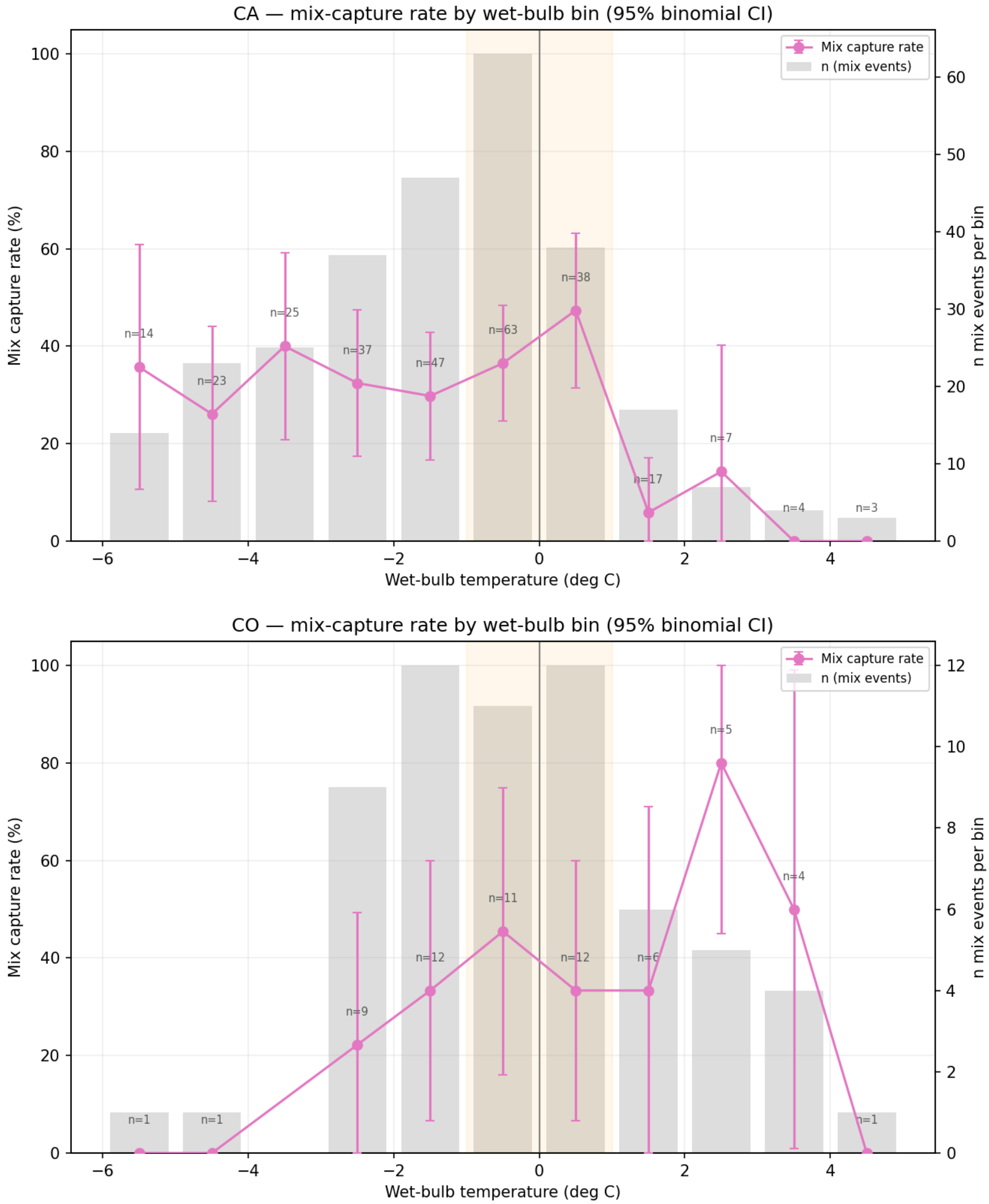


Fig. A9. Mix-capture rate by 1 °C wet-bulb bin for California (top) and Colorado (bottom), held-out test split. Points give the fraction of observer-reported mix events per bin whose calibrated $p\,(\text{snow})$ falls inside the uncertainty band, with 95% intervals; grey bars give the mix-event count per bin on the right-hand axis, and shading marks $|T_w| \leq 1\,^\circ\text{C}$. Two results stand out. Capture is highest in the near-freezing bins where mixed precipitation actually occurs (47% at $T_w = 0.5\,^\circ\text{C}$ in California) and falls away on either side, which is the intended behavior, since the band is placed to flag ambiguity rather than to maximize capture everywhere. And pooled capture, 31% in California and 36% in Colorado, agrees with the independent test-set bootstrap estimates of 0.30 and 0.33 in Table 3. Interval construction differs from every other estimate in this paper. The saved prediction file for this diagnostic carries no cluster key, so the intervals shown are simple per-bin binomial intervals that treat mix events as independent. Because nearby simultaneous reports are not independent, these are optimistic, and the pooled figures in Table 3 are the ones to quote. Several tail bins rest on fewer than ten events, where a rate of 0% or 100% reflects a handful of observations; 42 California and 2 Colorado mix events fall outside the ±6 °C range and are absent here while remaining in the pooled totals.